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ECM617 - RADIO FREQUENCY SYSTEM DESIGN

TECHNICAL REPORT: LNA DESIGN

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1.0 ABSTRACT

The rapid expansion of sub-6 GHz 5G networks has intensified the need for highly linear, low-noise RF front-end components capable of operating efficiently within densely packed spectrum environments. Among these, the Low-Noise Amplifier (LNA) plays a pivotal role, serving as the first active stage in the receiver chain and dictating the overall system sensitivity through its gain, noise performance, matching accuracy, and stability. This design specifically targets key performance metrics such as the Stability Factor (K), Voltage Standing Wave Ratio (VSWR), forward gain (S21), and Noise Figure (NF), as these parameters collectively determine the robustness and sensitivity of the amplifier in practical 5G deployments.

This report presents the complete design and evaluation of a 3.6 GHz LNA employing the QPL9057 transistor, a broadband, low-noise gain block suitable for modern wireless infrastructure. The design integrates lumped-element matching, distributed microstrip implementation, and optimization using Keysight Genesys. The methodology encompasses stability assessment, gain and noise analysis, Smith chart-based impedance matching, and systematic transformation of lumped components into distributed structures suitable for high-frequency fabrication. Simulation results before and after optimization demonstrate significant improvements in gain, matching, and noise performance. The final design provides a stable, high-gain solution suitable for 5G receiver applications, highlighting the interplay between device characteristics, matching topology, and transmission-line-based implementation on a realistic substrate.

2.0 INTRODUCTION

The deployment of 5G New Radio (NR) has introduced stringent requirements for receiver sensitivity, linearity, and noise performance, particularly within the sub-6 GHz spectrum where most early 5G deployments occur. The 3.5–3.7 GHz range, commonly referred to as the mid-band or C-band, offers an advantageous balance between coverage and capacity, making it one of the most critical frequency allocations for nationwide 5G infrastructure. In this context, the Low-Noise Amplifier (LNA) serves as a fundamental building block in the receiver front-end, responsible for ensuring that extremely weak incoming signals are amplified with minimal degradation.

The LNA's importance arises from its position as the first active stage after the antenna, where its performance directly dictates the overall sensitivity according to the Friis noise formula. A well-designed LNA must achieve high gain, low noise, and robust stability without introducing excessive distortion. For 5G applications, these requirements become more challenging, as the operating frequencies are higher and the spectrum is more congested, making impedance matching and stability control more demanding. Consequently, the selection of an appropriate device and the design of optimal matching networks are crucial to achieving the targeted specifications.

In this work, the QPL9057 transistor is selected as the core active device for a 3.6 GHz LNA. This transistor exhibits wideband gain characteristics, low intrinsic noise, and broadband internal matching, making it suitable for mid-band 5G receiver architectures. To ensure the amplifier delivers consistent and reliable performance within this band, the design process focuses on optimizing key parameters such as the Stability Factor (K), Voltage Standing Wave Ratio (VSWR), forward gain (S21), and Noise Figure (NF). These metrics collectively determine the amplifier's ability to operate safely, efficiently, and sensitively under practical signal conditions.

Designing at 3.6 GHz also introduces practical engineering challenges. At this frequency, the influence of parasitic inductances and capacitances becomes significant, necessitating precise modeling of both the active device and passive components. Smith chart techniques are essential for visualizing impedance transformations and selecting appropriate matching topologies. Additionally, while lumped components provide conceptual clarity and initial validation, transitioning the design into a realizable microwave circuit requires the use of distributed transmission-line elements that account for substrate properties, physical dimensions, and electromagnetic behavior.

To address these challenges, this report presents a complete workflow comprising S-parameter extraction, analytical calculations, Smith chart-based matching, lumped-to-distributed transformation, and performance optimization using Keysight Genesys. Simulation results—both before and after network refinement demonstrate how targeted adjustments enable improvements in gain, matching accuracy, noise figure, and stability. Through this systematic approach, the final LNA design achieves performance suitable for 5G mid-band receivers, while also illustrating the interplay between theory, simulation, and practical implementation in modern RF circuit engineering.

3.0 METHODOLOGY

This section describes the analytical and simulation-based methods used in developing the 3.6 GHz LNA using the QPL9057 transistor. The methodology integrates S-parameter extraction, stability evaluation, gain analysis, matching network synthesis, and subsequent transformation into distributed microstrip elements suitable for high-frequency implementation. Every calculation and design decision is grounded in RF amplifier theory to ensure that the final design satisfies the performance objectives of mid-band 5G receivers.

3.1 Chosen Transistor

The QPL9057 transistor, manufactured by Qorvo, is selected as the active device for this 3.6 GHz LNA design. It is a broadband, low-noise gain block that integrates internal matching networks optimized across a wide frequency range, making it suitable for emerging 5G sub-6 GHz infrastructures. Its inherent characteristics such as low minimum noise figure, relatively high gain, and stable broadband behavior provide a strong foundation for achieving the targeted LNA performance.

Although the QPL9057 includes internal matching, external networks remain essential to tailor the amplifier specifically to 3.6 GHz. These external networks refine input and output impedance, control gain, and ensure stability under varying source and load conditions. The transistor's S-parameters at the design frequency were extracted from the uploaded S2P file and serve as the fundamental dataset for all subsequent analytical steps.

QORVO

QPL9057
Ultra-Low Noise Flat Gain Amplifier

S-Parameters

Test Conditions: V_{DD}=+5 V, I_{DD}=50 mA (typ.), T=+25°C, Reference planes at device pins

Freq (GHz)	S11 (mag)	S11 (ang)	S21 (mag)	S21 (ang)	S12 (mag)	S12 (ang)	S22 (mag)	S22 (ang)
1.0	0.363	-62.40	12.256	40.27	0.021	-7.21	0.079	-111.83
1.1	0.349	-64.72	11.937	31.80	0.021	-11.15	0.057	-122.11
1.2	0.338	-66.97	11.726	23.70	0.021	-15.27	0.037	-134.96
1.3	0.333	-69.47	11.590	15.99	0.021	-19.39	0.022	-155.81
1.4	0.331	-72.56	11.536	8.54	0.021	-23.49	0.011	153.14
1.5	0.330	-75.77	11.540	1.25	0.021	-27.66	0.016	86.83
1.6	0.332	-79.54	11.620	-5.84	0.020	-31.88	0.025	61.12
1.7	0.337	-83.50	11.751	-12.88	0.020	-36.27	0.034	47.76
1.8	0.345	-88.04	11.924	-19.83	0.020	-40.95	0.042	34.14
1.9	0.354	-92.89	12.173	-26.82	0.019	-45.86	0.050	24.15
2.0	0.365	-98.45	12.460	-33.85	0.019	-51.11	0.054	12.78
2.1	0.376	-104.44	12.798	-41.05	0.019	-56.72	0.058	-1.99
2.2	0.390	-110.49	13.197	-48.32	0.018	-62.58	0.065	-16.74
2.3	0.403	-117.36	13.609	-55.92	0.018	-68.97	0.070	-31.21
2.4	0.420	-124.54	14.058	-63.77	0.017	-76.11	0.077	-47.98
2.5	0.434	-132.38	14.508	-71.91	0.017	-83.90	0.090	-65.73
2.6	0.449	-140.69	14.994	-80.39	0.016	-92.51	0.107	-82.09
2.7	0.463	-149.41	15.403	-89.27	0.016	-101.95	0.126	-97.44
2.8	0.473	-158.73	15.747	-98.57	0.015	-112.06	0.150	-112.41
2.9	0.480	-168.68	15.976	-108.18	0.014	-123.37	0.177	-126.71
3.0	0.481	-179.03	16.074	-118.07	0.014	-135.08	0.210	-139.42
3.1	0.478	170.34	15.996	-128.17	0.014	-147.69	0.242	-151.03
3.2	0.467	159.39	15.696	-138.34	0.013	-160.17	0.270	-162.97
3.3	0.450	148.27	15.228	-148.37	0.013	-172.30	0.300	-174.34
3.4	0.428	137.38	14.633	-158.07	0.013	175.93	0.327	176.01
3.5	0.404	127.03	13.903	-167.45	0.013	165.17	0.347	166.69
3.6	0.381	116.60	13.101	-176.23	0.013	155.28	0.364	157.49
3.7	0.357	106.24	12.271	175.47	0.013	146.87	0.378	149.10
3.8	0.332	96.18	11.474	167.75	0.013	139.24	0.386	141.83
3.9	0.314	83.15	10.730	160.14	0.013	132.16	0.386	134.10
4.0	0.296	74.96	9.999	153.51	0.014	127.06	0.392	128.18
4.1	0.282	66.96	9.308	147.28	0.014	122.02	0.396	122.56
4.2	0.267	58.82	8.687	141.49	0.014	118.13	0.400	117.01
4.3	0.252	51.86	8.143	136.02	0.015	114.50	0.403	112.77
4.4	0.238	45.26	7.647	130.75	0.015	110.83	0.401	108.89
4.5	0.227	39.31	7.182	125.74	0.016	107.90	0.401	104.56
4.6	0.216	32.68	6.770	120.86	0.016	105.34	0.402	100.53
4.7	0.203	26.93	6.392	116.21	0.017	102.88	0.400	96.95
4.8	0.190	20.82	6.056	111.63	0.018	100.30	0.398	93.19
4.9	0.178	15.55	5.740	107.27	0.018	97.92	0.397	89.45
5.0	0.169	10.06	5.448	103.06	0.019	95.81	0.396	86.01
5.1	0.156	4.12	5.181	98.77	0.020	93.45	0.394	82.07
5.2	0.141	-1.67	4.943	94.71	0.020	91.54	0.394	78.96
5.3	0.132	-7.43	4.722	90.73	0.021	89.25	0.391	75.43
5.4	0.122	-14.31	4.515	86.71	0.022	87.22	0.388	71.86
5.5	0.112	-22.69	4.315	82.73	0.023	84.51	0.388	68.09
5.6	0.102	-32.31	4.133	78.92	0.024	82.03	0.388	65.06
5.7	0.093	-41.13	3.971	75.14	0.025	79.79	0.385	61.53
5.8	0.090	-51.19	3.818	71.35	0.025	77.65	0.389	57.47
5.9	0.089	-63.48	3.666	67.57	0.026	75.64	0.389	54.11
6.0	0.091	-76.48	3.537	63.84	0.027	73.59	0.387	50.63

Datasheet, Rev. G, January 30, 2024 | Subject to change without notice 3 of 12 www.qorvo.com

Figure 1: QPL9057 S-Parameters

QORVO

QPL9057
Ultra-Low Noise Flat Gain Amplifier

Noise Parameters

Test conditions: V_{DD}= +5 V, I_{DD}=50 mA (typ.), T= +25°C, Reference planes at device pins

Freq (GHz)	NF_{min} (dB)	Gamma Opt (mag)	Gamma Opt (deg)	Rn (Ω)
2.0	0.246	0.283	47.66	0.076
2.2	0.261	0.231	68.48	0.061
2.4	0.279	0.213	81.15	0.054
2.6	0.315	0.198	92.03	0.050
2.8	0.334	0.191	103.07	0.046
3.0	0.408	0.146	110.96	0.053
3.2	0.413	0.189	117.92	0.044
3.4	0.450	0.162	131.84	0.045
3.5	0.491	0.131	143.21	0.048
3.6	0.531	0.157	152.91	0.043
4.0	0.599	0.099	172.03	0.050
4.2	0.634	0.142	-176.97	0.051
4.4	0.686	0.142	-160.26	0.051
4.6	0.681	0.149	-153.24	0.057
4.8	0.775	0.102	-141.49	0.069
5.0	0.817	0.138	-132.51	0.076
5.2	0.863	0.149	-131.67	0.079
5.4	0.957	0.134	-130.72	0.092
5.6	1.051	0.185	-114.16	0.107
5.8	1.184	0.178	-113.36	0.109

Figure 2: QPL9057 Noise Parameters

3.2 Calculations

The design process begins by analyzing the S-parameters of the QPL9057 at 3.6 GHz to establish the amplifier's baseline behavior. These parameters inform the calculations of maximum gain, stability metrics, and impedance conditions required for effective matching.

$S_{11} = 0.381 \angle 116.6$, $S_{21} = 13.101 \angle -176.23$, $S_{12} = 0.013 \angle 155.28$, $S_{22} = 0.364 \angle 157.49$

3.2.1 Find MSG

A crucial parameter utilized in the design of Low Noise Amplifiers (LNAs) is Maximum Stable Gain (MSG), which establishes the maximum gain a transistor can deliver before become unstable. Because feedback effects can cause transistors to become unstable at high frequencies, MSG is essential for maintaining dependable operation.

a) $MSG = \frac{S_{21}}{S_{12}} = \frac{13.101}{0.013} = 1007.8$

In dB $= 10 \log_{10} 1007.8 = 30\,dB$

3.2.2 Find G_{lmax} and G_{smax}

$G_{L\,max} = \frac{1}{1 - S_{22}^2} =$ [redacted] $1 - 0.364$ Radius, SC

$= 1.153 = 0.618\,dB$

$G_{s\,max} = \frac{1}{1 - S_{11}^2} = \frac{1}{1 - 0.381^2} = 1.17 = 0.682\,dB$

3.2.3 Find G_{Tumax}

$G_{TU\,max}(dB) = G_{s\,max} + G_0 + G_{l\,max}$, $G_0 = 10 \log_{10} S_{21}^2$

$= 0.682 + 22.35 + 0.618$ $\quad = 10 \log_{10} 13.101^2$

$= 23.65\,dB$ $\quad = 22.35\,dB$

3.2.4 Find k

$$k = \frac{1 + |\Delta|^2 - |S_{11}|^2 - |S_{22}|^2}{2|S_{12}||S_{21}|}$$

$$= \frac{1 + 0.168^2 - 0.781^2 - 0.364^2}{2(13.101)(0.013)}$$

$$= 2.2$$

$$\Delta = S_{11}S_{22} - S_{12}S_{21}$$

$$= 0.139\angle -85.91 - 0.17\angle -20.95$$

$$= 0.168\angle -152.4$$

3.2.5 Find input stability

$$C_S = (S_{11} - \Delta S_{22}^*)^* \left[|S_{11}|^2 - |\Delta|^2\right]^{-1}$$

$$= 0.361\angle -125.54\,[8.55]$$

$$= 3.09\angle -125.54$$

$$= 3.09 \times 8.25 = 25.5$$

$$r_S = \left|\frac{S_{12}S_{21}}{|S_{11}|^2 - |\Delta|^2}\right|$$

$$= \left|\frac{0.17\angle -20.95}{0.117}\right|$$

$$= 1.453 \times 8.25 = 11.99$$

3.2.6 Find output stability

$$C_L = (\Delta S_{11}^* - S_{22})^* \left[|\Delta|^2 - |S_{22}|^2\right]^{-1}$$

$$= (0.344\angle 12.67)[-9.59]$$

$$= 3.3\angle -167.33$$

$$= 3.3 \times 8.25 = 27.23$$

$$r_L = \left|\frac{S_{12}S_{21}}{|S_{22}|^2 - |\Delta|^2}\right|$$

$$= \frac{0.17\angle -20.95}{0.104}$$

$$= 1.635 \times 8.25$$

$$= 13.49$$

3.2.7 Unilateral Approximation

$G_{TUmax}(dB) = 23.65\ dB \approx 20\ dB$

Gain = 25dB

transistor I/P O/P

17dB 2.5dB 0.5dB

$G_s = 2.5dB = 1.78$ 1.25

$g_s = \frac{G_s}{G_{s\,max}} = \frac{1.78}{1.17} = 1.52 = 1.82\ dB$ $1.07 = 0.3dB$

a) For 2.8 db input gain

$G_s = 2.5dB = 1.78$ 1.25

$g_s = \frac{G_s}{G_{s\,max}} = \frac{1.78}{1.17} = 1.52 = 1.82\,dB$ 1.07 = 0.3dB

center

$$d_s = \frac{g_s|S_{11}|}{1-|S_{11}|^2(1-g_s)} = \frac{1.52\times0.381}{1-(0.381^2)(1-1.52)} = 0.54\times8.25 = 4.5$$

radius

$$R_s = \frac{\sqrt{1-g_s}(1-|S_{11}|^2)}{1-|S_{11}|^2(1-g_s)} = \frac{\sqrt{1-1.52}(1-0.381^2)}{1-(0.381)^2(1-1.52)} = \frac{0.6164}{0.9245} = 0.67\times8.25 = 5.5$$

b) For 1.0 db noise

1dB noise circle

$F_i = 1dB = 1.259$

$F_{min} = 0.53 = 1.13$

$\Gamma_o = 0.157\angle152.91$

$$N_i = \frac{F_i - F_{min}}{4r_n}|1+\Gamma_o|^2 = \frac{1.259-1.13}{4\times0.043}|1+(-0.14+j0.071)|^2$$

$$= 0.9675+j0.1215$$

$$= 0.98\angle7.16 \times 8.25$$

$$= 8.1$$

Center $C_{F1} = \frac{\Gamma_o}{1+N_i} = \frac{0.157\angle152.91}{1+0.98} = 0.08\angle152.91 = 0.08\times8.25 = 0.66$

radius $R_{F1} = \frac{1}{1+N_i}\sqrt{N_i^2+N_i(1-|\Gamma_o|^2)} = \frac{1}{1+0.98}\sqrt{(0.98)^2+0.98(1-|0.157|^2)}$

$$= 0.7\times8.25 = 5.8$$

c) For 0.2 db output gain

0.5dB output gain

$G_1 = 0.5dB = 1.1$

$g_1 = \frac{G_1}{G_{max}} = \frac{1.1}{1.153} = 0.95$

center, $d_1 = \frac{g_1|S_{22}|}{1-|S_{22}|^2(1-g_1)} = \frac{0.95\times0.364}{1-(0.364)^2(1-0.95)} = 0.35\times8.25 = 2.9$

radius, $R_1 = \frac{\sqrt{1-g_1}(1-|S_{22}|^2)}{1-|S_{22}|^2(1-g_1)} = \frac{\sqrt{1-0.95}(1-0.364^2)}{1-0.364^2(1-0.95)} = 0.195\times8.25 = 1.6$

3.2.8 Smith Chart Plot

3.2.8.1 Input Stability

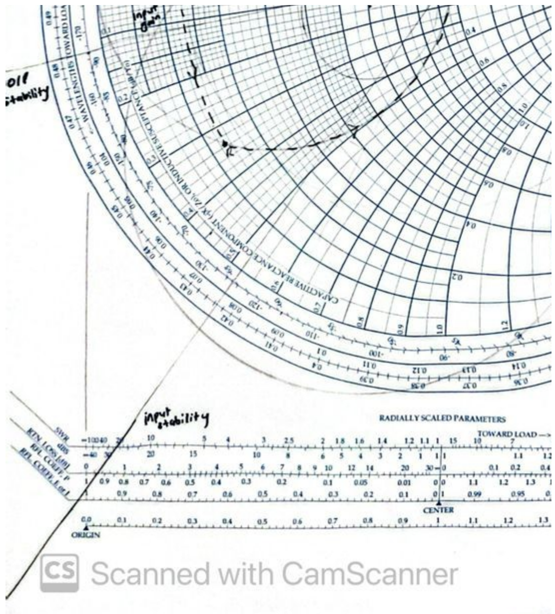


Figure 3: Input Stability Circle

In Figure 3, the radius of the input stability circle is sufficiently large to lay completely outside the Smith chart, demonstrating unconditional stability over the whole chart.

3.2.8.2 Output Stability

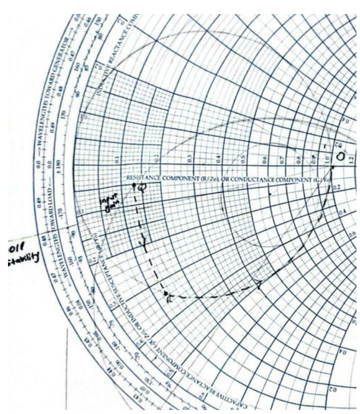


Figure 4: Output Stability Circle

In Figure 4, the radius of the output stability circle is sufficiently large to lay completely outside the Smith chart, showing unconditional stability throughout.

3.2.9 Input Matching Network

Reactance calculations were employed to design the input matching network:

a) Capacitance

Point Q on the noise circle was selected as it represents the farthest stable point from the input stability circle while maintaining intersection with the input gain circle. The coordinate transformation from point Q to point R in anticlockwise was then calculated:

$Q \rightarrow R$

$P(1.18 - j1.7)$

$Q(0.13 - j0.17)$

$R(0.13 - j0.33)$

$R - Q$ (series capacitor)

$(0.13 - j0.33) - (0.13 - j0.17)$

$= -j0.16 = jx$

$jX = jx \times 50 = -j8\,\Omega$

$X_C = \frac{-j}{wC}$

$-j8 = \frac{-j}{wC}$

$C = \frac{1}{w8} = \frac{1}{2\pi \times 3.6 \times 10^9 \times 8} = 5.53\,pF$

Q to R move in anticlockwise resulting in series capacitor reactance X_{c}.

b) Inductance

R→O

R to O move in anticlockwise resulting in parallel inductance reactance X_L

$R \rightarrow O$ (parallel inductor)

$R(1+j2.7) \quad O(1+j0)$

$(1+j0)-(1+j2.7) = -j2.7 = jb$

$jB = jb \div 50 = -j0.05$

$X_L = \frac{-j}{wL}, \quad \frac{-j}{wL} = -j0.05$

$L = \frac{1}{w0.05} = \frac{1}{2\pi \times 3.6 \times 10^9 \times 0.05} = 0.88nH$

3.2.10 Output Matching Network

Reactance calculations were employed to design the output matching network:

a) Capacitance

$S \rightarrow T$

S to T move in anticlockwise resulting in series capacitor reactance X_c

$S \rightarrow T$ (series capacitor)

$S(0.75 - j0.15)$, $T(0.75 - j0.43)$

$(0.75 - j0.15) - (0.75 - j0.43) = j0.28 = jb$

$jB = jb \times 50 = j14\,\Omega$

$C = \dfrac{jB}{j\omega} = \dfrac{14}{2\pi \times 3.6^{9}} = 0.74\,nF$

b) Inductance

$T \rightarrow O$

T to O move in anticlockwise resulting in parallel inductance reactance X_L

$T \rightarrow O$ (parallel inductor)

$T(1 - j0.58)$, $O(1 + j0)$

$(1 + j0) - (1 - j0.58) = j0.58 = jx$

$jX = jx \div 50 = j0.0116$

$L = \dfrac{jx}{j\omega} = \dfrac{0.0116}{2\pi \times 3.6 \times 10^{9}} = 0.62\,nH$

4.0 RESULT AND DISCUSSIONS

4.1 Simulation Result

4.1.1 Lumped Circuit Before Optimization

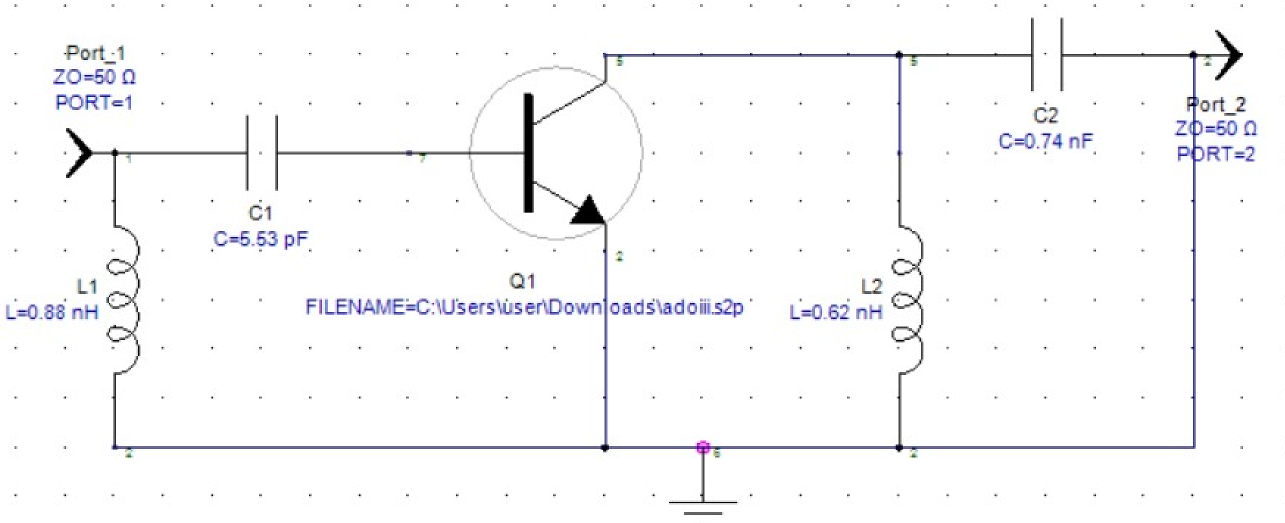


Figure 5: Lumped Circuit Layout Before Optimization

The initial lumped-element LNA design is shown in Figure 5. The circuit consists of the QPL9057 transistor connected between two 50-Ω ports, with input and output matching networks implemented using discrete inductors and capacitors. At the input, the matching network is formed by a series capacitor C1 = 5.53 pF followed by a shunt inductor L1 = 0.88 nH, which transforms the source impedance toward the required input impedance of the transistor at 3.6 GHz. The transistor model is loaded directly from the measured S-parameter file provided.

On the output side, a series capacitor C2 = 0.74 nF and a shunt inductor L2 = 0.62 nH provide the initial impedance transformation toward the 50-Ω load at Port 2. The transistor is referenced to ground at its source terminal, and both ports are defined with characteristic impedances of 50 Ω to emulate standard RF system conditions.

This initial lumped circuit represents the first-pass design based purely on analytical matching calculations, serving as the starting point for further refinement and optimization.

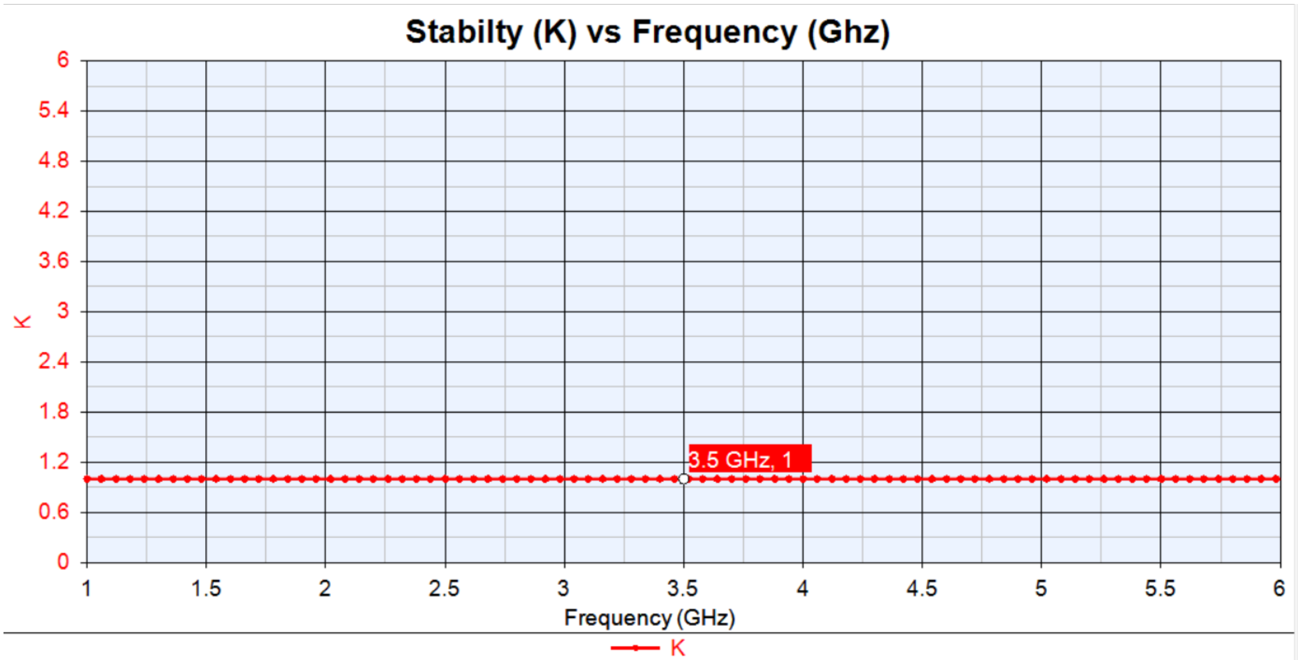


Figure 6: Graph Of Stability K Before Optimized

The stability plot in Figure 6 shows the Rollett stability factor K for the initial lumped LNA design across the frequency range of 1–6 GHz. The curve remains relatively flat and stays close to K = 1 throughout the entire band. At the design frequency of 3.6 GHz, the circuit achieves a K-factor of approximately 1.0, which indicates that the amplifier is only marginally stable.

Because unconditional stability requires $K > 1$ at all frequencies, the initial circuit does not provide sufficient stability margin. Although the amplifier is not unstable, the low K-factor suggests that small changes in component tolerances, transistor variations, or layout parasitic could push the circuit into a potentially unstable region. This highlights the need for further optimization, particularly in adjusting the matching network or adding stabilizing elements, to ensure robust operation across the intended operating band.

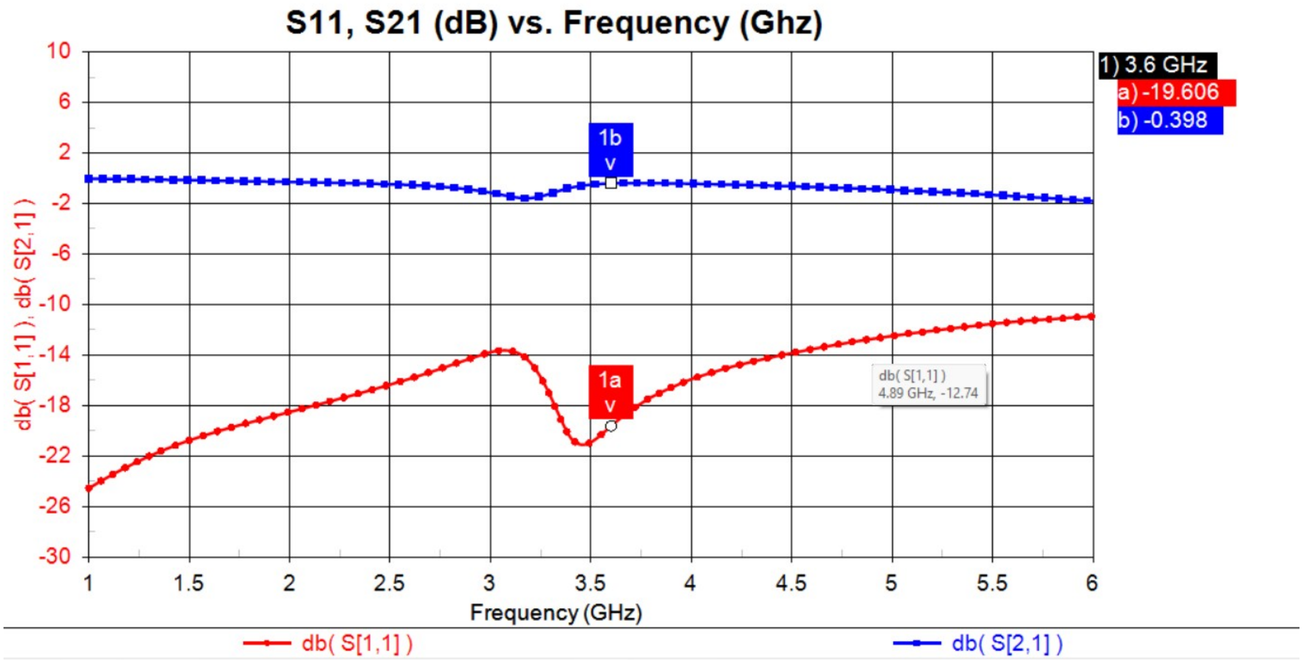


Figure 7: Graph of Gain Before Optimized

The gain response of the initial lumped LNA design is shown in Figure 7. The forward gain remains relatively low across the entire frequency range, with a value of approximately –0.20 dB at 3.6 GHz. This indicates that the amplifier provides almost no gain, and in fact exhibits slight attenuation at the design frequency. Ideally, a low-noise amplifier should provide positive gain, typically in the range of 10–20 dB for practical receiver applications. The low value observed here confirms that the first-pass matching network is not effectively transforming the source and load impedances to their required optimum points.

The shape of the gain curve suggests that the transistor is not being presented with the correct input and output impedance conditions. This is expected, as the initial lumped matching network is based only on theoretical calculations and does not account for high-frequency parasitic or the frequency-dependent behaviour of the QPL9057 device. The mismatch between the designed matching point and the actual S-parameter behaviour of the transistor results in an inefficient power transfer, explaining the low gain.

Overall, the pre-optimization response highlights the need for refinement of the matching network to ensure proper power transfer and achieve the expected forward gain from the QPL9057. Subsequent optimization will focus on adjusting the component values to move the operating point toward the desired gain and performance targets at 3.6 GHz.

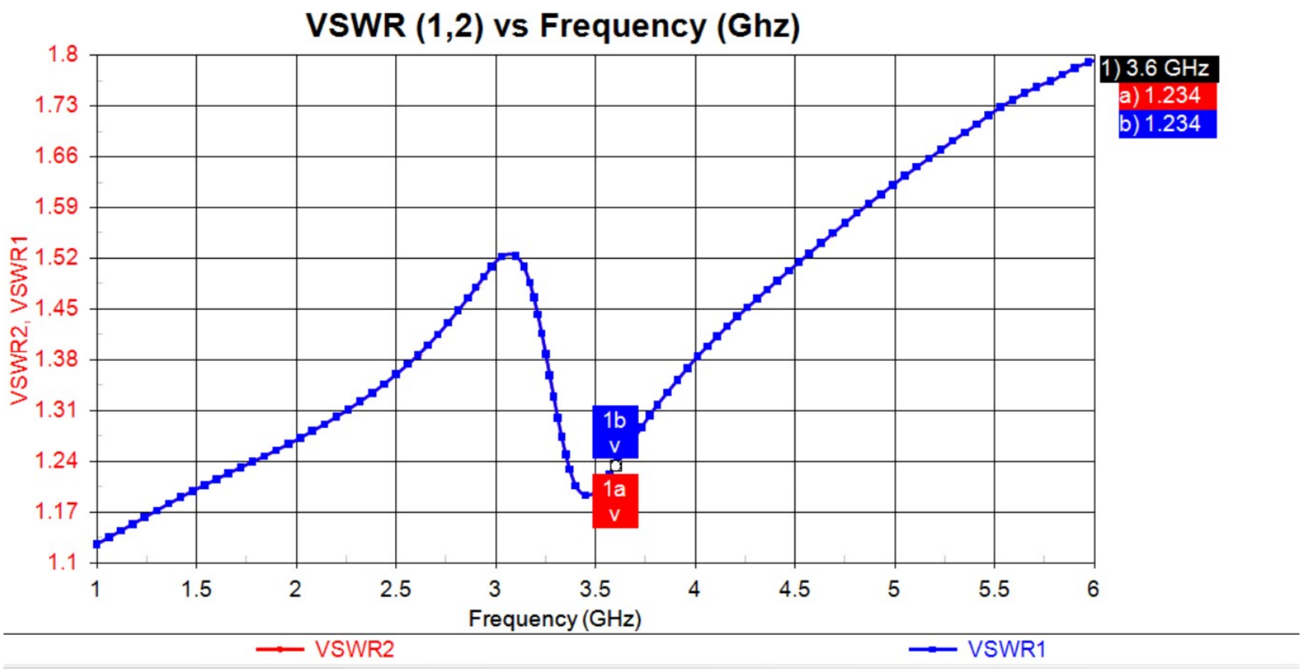


Figure 8: Graph of VSWR Before Optimized

As shown in Figure 8, the graph indicates that the VSWR1 and VSWR2 value is 1.234, representing the input VSWR1, while representing the output VSWR2. At 3.6 GHz, the circuit shows a VSWR of 1.234 at both the input and output ports. This value indicates that the matching between the amplifier and the 50-Ω system is reasonably good, with only a small portion of the signal being reflected back toward the source or load. A VSWR close to 1 means that the impedance seen by the ports is close to 50 Ω, and in this case the value of 1.234 shows that the initial matching network is already performing fairly well. However, even though the VSWR is acceptable, the overall amplifier gain is still low, which means that proper impedance matching alone is not enough and further optimization is needed to improve the LNA's performance.

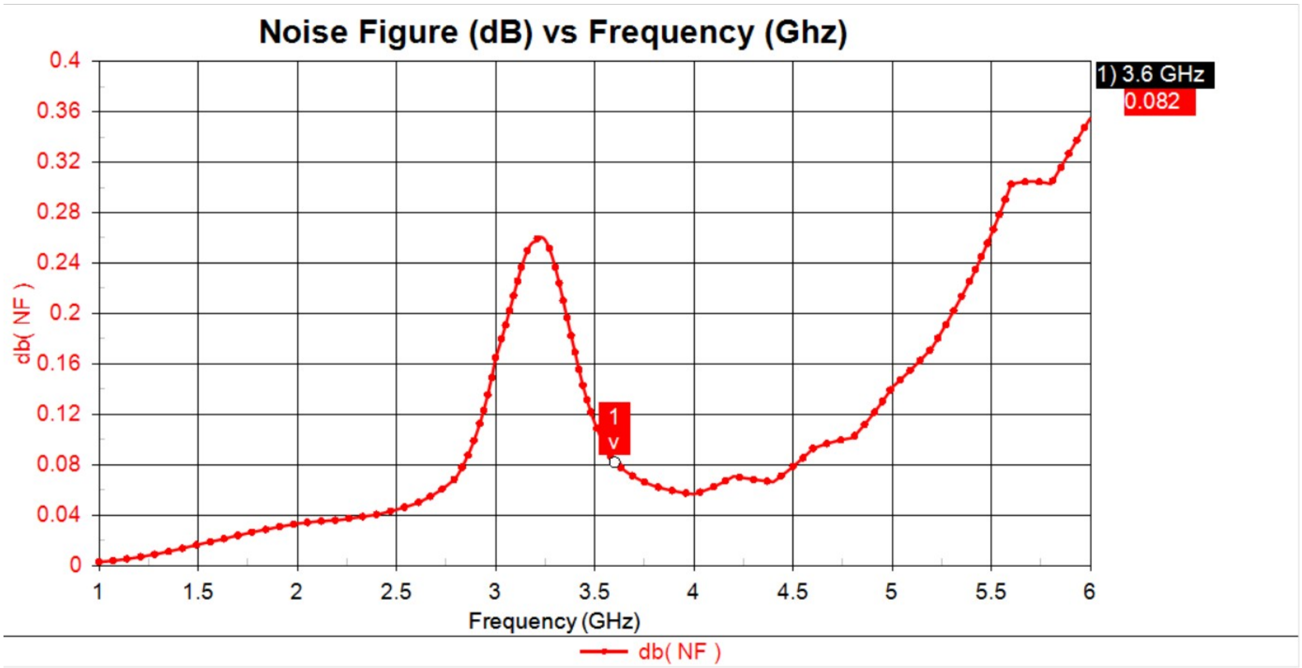


Figure 9: Graph of Noise Figure Before Optimized

Figure 9 shows the Noise Figure graph. The Noise Figure (NF) of the initial design at 3.6 GHz is 0.082, which indicates that the amplifier adds very little noise to the received signal. A lower NF is always desirable for an LNA because it helps maintain good receiver sensitivity, especially when dealing with weak signals. An NF value this low shows that the QPL9057 transistor is naturally low noise at the design frequency, even before any optimization is applied. However, although the noise performance is excellent, the overall amplifier gain is still insufficient, meaning further tuning of the matching network is required to achieve proper LNA performance while maintaining the low NF.

4.1.2 Lumped Circuit After Optimization

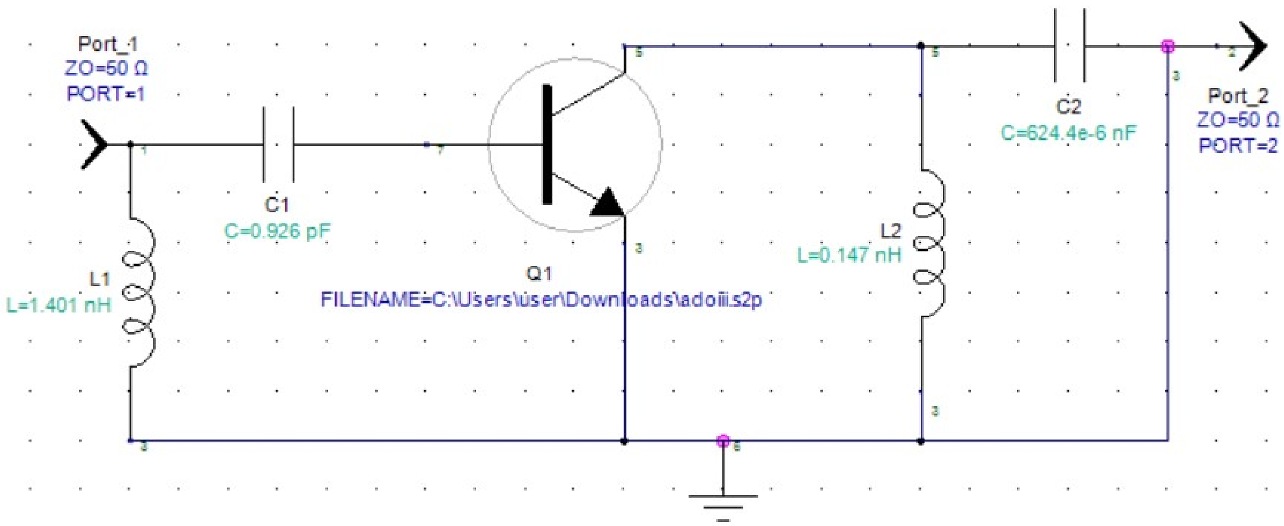
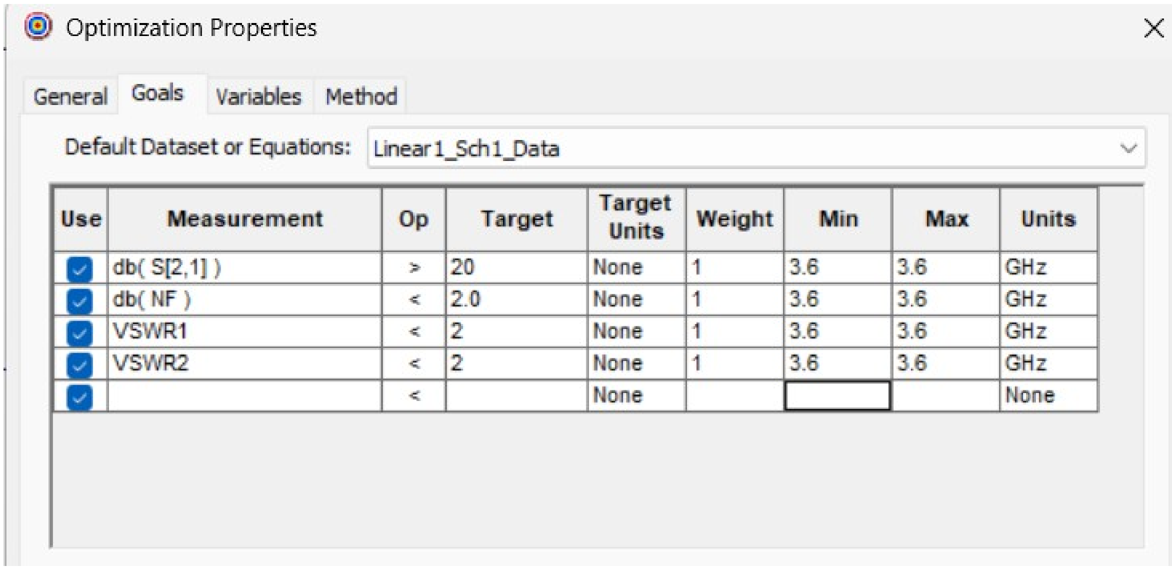


Figure 10: Lumped Circuit Layout After Optimization

Figure 10 displays the simulation results of the optimized circuit obtained from Genesys 2023. The input matching network is implemented using a 1.401 nH shunt inductor together with a 0.926 pF series capacitor, forming an LC structure that matches the 50 Ω source impedance to the transistor's input impedance at the operating frequency. This arrangement mitigates reactive effects at the input, leading to reduced signal reflection and improved power transfer. At the output, impedance matching is achieved by a 624.4 nF series capacitor in combination with a 0.147 nH shunt inductor, which adapts the transistor's output impedance to the 50 Ω load. All inductance and capacitance values were obtained through an iterative simulation-based optimization process aimed at minimizing return loss and maximizing power transfer efficiency.



Use	Measurement	Op	Target	Target Units	Weight	Min	Max	Units
☑	db(S[2,1])	>	20	None	1	3.6	3.6	GHz
☑	db(NF)	<	2.0	None	1	3.6	3.6	GHz
☑	VSWR1	<	2	None	1	3.6	3.6	GHz
☑	VSWR2	<	2	None	1	3.6	3.6	GHz
☑		<		None				None

Figure 11: Goals Of Optimization

Figure 11 presents the optimization objectives for the key circuit performance parameters. The gain (S21) is targeted to exceed 20 dB to ensure sufficient signal amplification, while both input and output VSWR are constrained to remain below 2 to achieve proper impedance matching. In addition, the noise figure is limited to less than 2 dB to minimize noise contribution and enhance overall circuit performance. These requirements follow the coursework guidelines to obtain optimal values for noise figure, gain, and VSWR. The stability factor (K) is not included in the optimization goals, as it remains unchanged and already indicates unconditional stability with a value of K = 1. Collectively, these targets define the criteria used to achieve the best overall circuit performance.

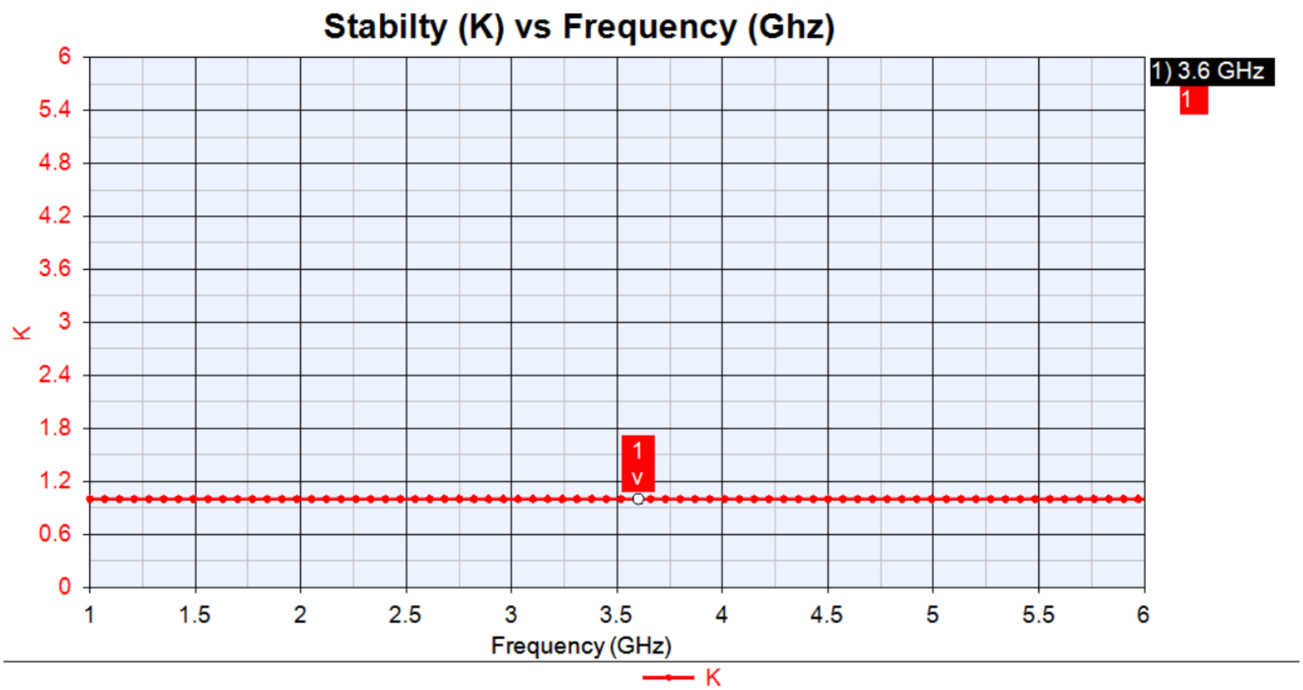


Figure 12: Graph Of Stability After Optimized

Figure 12 illustrates the post-optimization stability analysis, showing that the stability factor (K) remains at **1**. This value is unchanged from the original lumped circuit, as the design was already unconditionally stable prior to optimization. Consequently, the stability factor was not included as an optimization objective, allowing the optimization process to focus solely on improving other critical performance parameters such as gain, noise figure and VSWR while maintaining the existing stable operatingcondition.

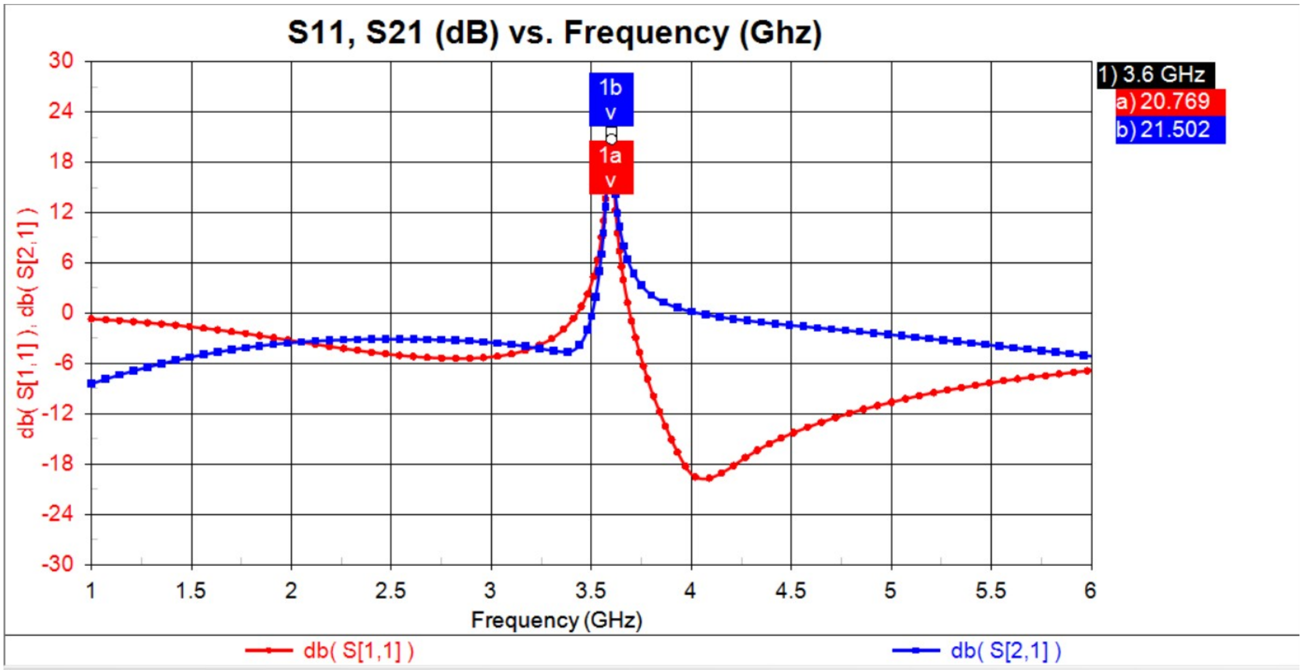


Figure 13: Graph of Gain After Optimize

Figure 13 presents the optimized gain performance, achieving a gain of 21.502 dB, which is a clear improvement over the pre-optimization results and substantially exceeds the minimum design requirement of 20 dB. This enhanced gain demonstrates the effectiveness of the optimization process in significantly improving the amplifier's amplification capability. At the same time, the achieved result indicates that the gain enhancement was obtained without compromising other critical performance parameters, reflecting a well-balanced design that ensures stable and reliable overall system operation.

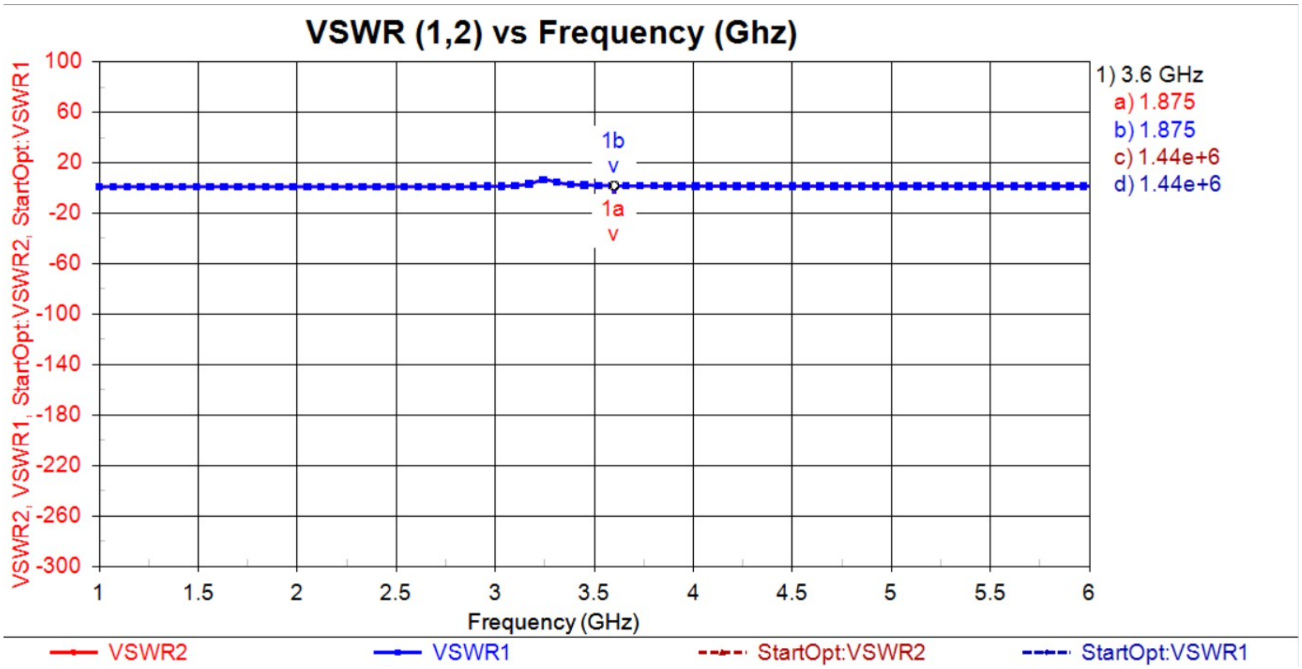


Figure 14: Graph of VSWR After Optimized

Shown in Figure 14, the input and output VSWR values after optimization are both approximately 1.875. Since both VSWR 1 and VSWR 2 have the same value, the ratio is low, indicating good impedance matching across both ports. Although this value is slightly higher than the ideal VSWR of 1, it remains below the target threshold of 2. This deliberate relaxation allowed improvements in other key performance metrics, including noise figure reduction and enhanced circuit stability. The results illustrate the trade-offs inherent in multi-objective LNA design, where slightly higher VSWR can enable superior overall system performance by balancing matching, noise, and stability.

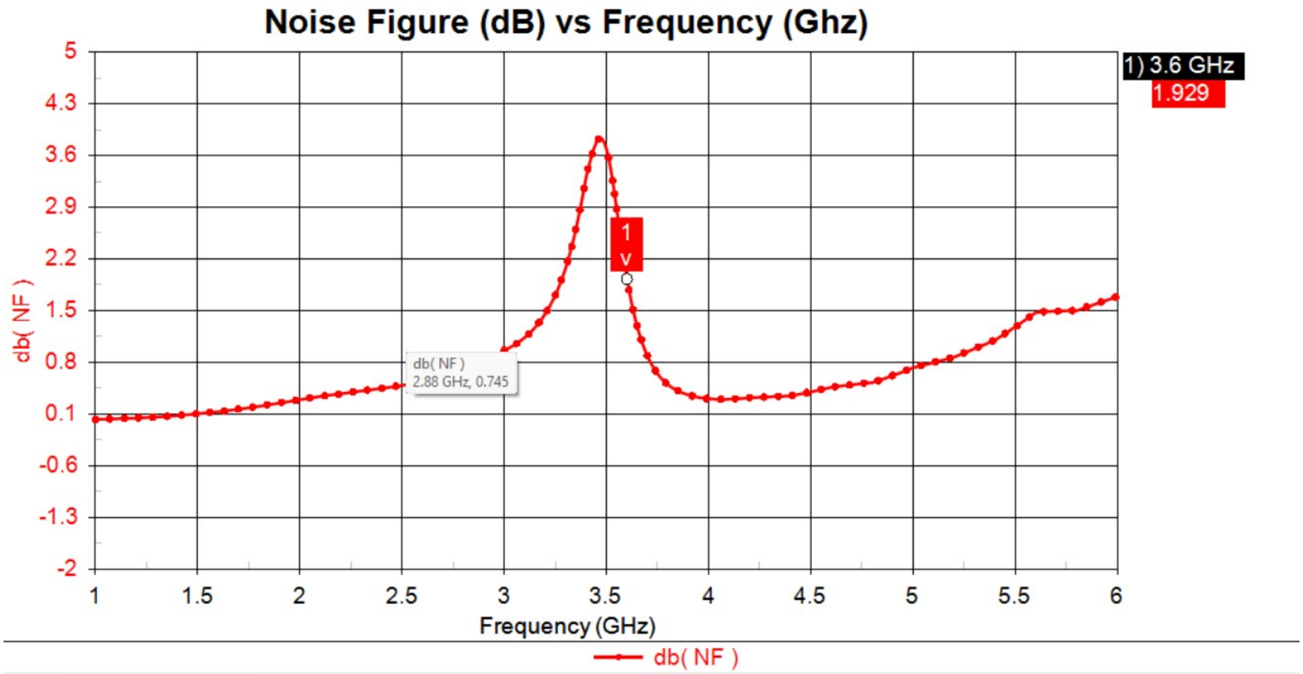


Figure 15: Graph of Noise Figure After Optimized

Figure 15 shows the post-optimization noise figure of the LNA, measured at 1.929 dB. This value meets the design requirement of NF < 2 dB, indicating that the amplifier introduces minimal noise to the system. A low noise figure is essential for maintaining high receiver sensitivity, as it determines the weakest signal level that can be reliably detected. By achieving this target, the design ensures effective performance in low-signal conditions while maintaining other critical parameters.

4.1.3 Lumped-To-Distributed Element Transformation

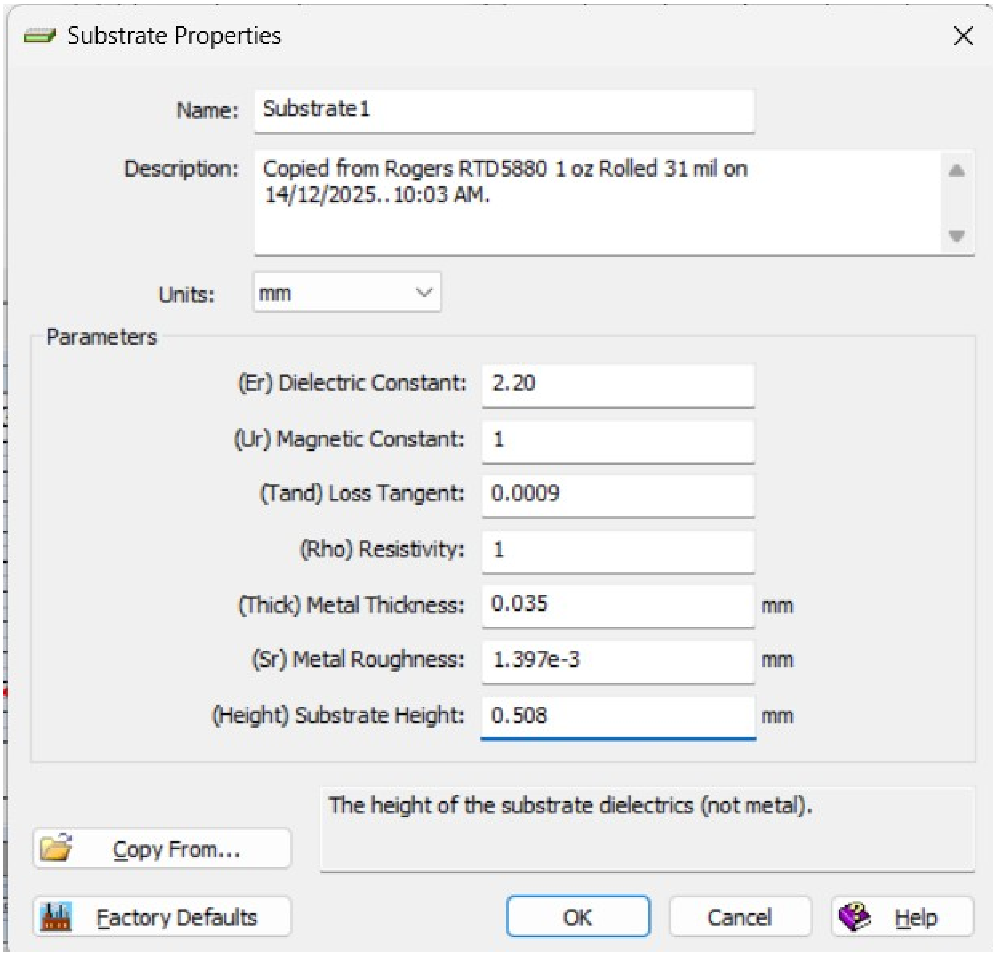


Figure 16: Substrate Properties

In distributed RF circuit design, signal propagation is treated as an electromagnetic wave phenomenon, where the physical dimensions of transmission lines are no longer negligible compared to the operating wavelength. Unlike lumped-element circuits that assume instantaneous signal behaviour, distributed circuits explicitly account for phase delay, propagation effects, and impedance variation along the transmission path, making them more suitable for microwave-frequency applications such as low-noise amplifiers.

For the distributed implementation of the LNA, Rogers RT/duroid 5880 is selected as the substrate with a thickness of 0.508 mm. The material has a relative dielectric constant ($\varepsilon r = 2.2$) and a relative permeability ($\mu r = 1$), ensuring stable and predictable electromagnetic behaviour. It's extremely low loss tangent ($\tan \delta = 0.0009$) minimizes dielectric losses at the operating frequency of 3.6 GHz, which is essential for preserving gain and maintaining a low noise figure.

The conductive layer is modelled with a metal thickness of 0.036 mm, representing a standard copper cladding suitable for high-frequency applications. A resistivity value ($\rho = 1$) is used to indicate efficient current conduction, while the surface roughness parameter ($Sr = 1.397 \times 10^{-3}$ mm) accounts for conductor loss effects caused by microscopic surface irregularities. These parameters collectively contribute to an accurate representation of conductor behaviour at microwave frequencies.

Overall, the selected substrate height of 0.508 mm, combined with its low-loss dielectric and well-defined conductor properties, provides accurate impedance control and reduced attenuation in the distributed matching network. This makes Rogers RT/duroid 5880 a suitable choice for implementing the microstrip-based LNA design at 3.6 GHz.

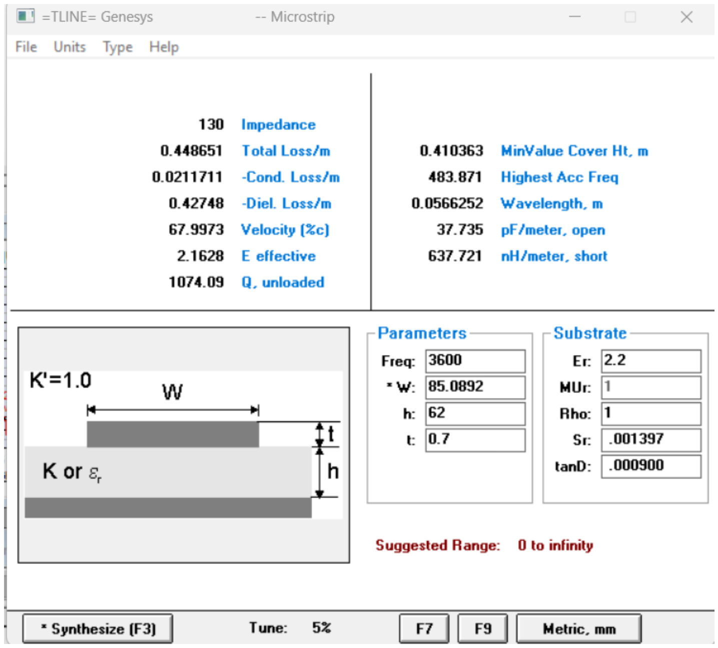


Figure 17: Syntheses Run Tline Of Microstrip Inductor

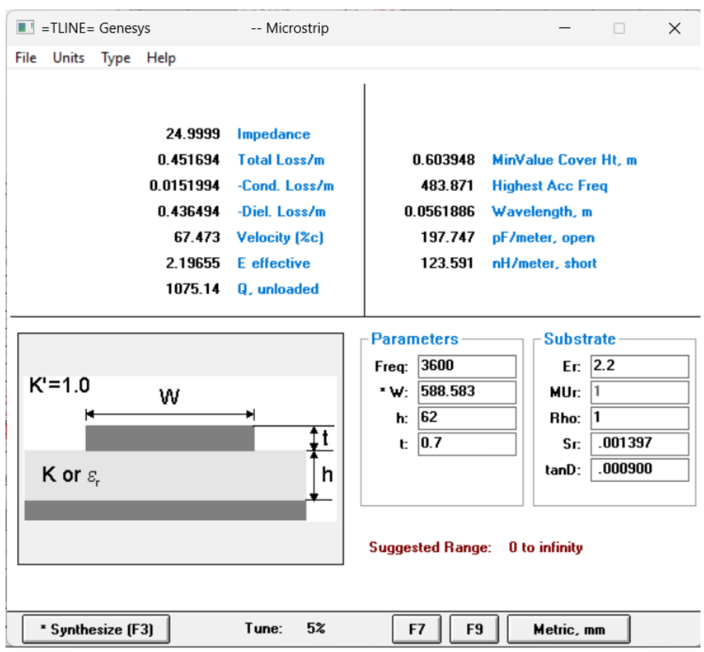


Figure 18: Syntheses Run Tline Of Microstrip Capacitor

Figures 17 and 18 present the microstrip configuration parameters used to realize lumped inductors and capacitors in their distributed equivalents. These figures specify key physical dimensions, particularly the microstrip line widths, which directly influence the effective inductance and capacitance achieved at the operating frequency. In addition, the effective dielectric constant (ε_eff) obtained from these design parameters allows the physical length of each microstrip section to be accurately determined using wavelength (λ) and transmission-line relationships. Through this approach, lumped components can be reliably translated into distributed structures while maintaining accurate impedance transformation and ensuring optimal circuit performance.

4.1.3.1 Calculations of lambda

$$f\sqrt{eff} \qquad 3.6G\sqrt{2.16585}$$

$$\lambda_L = \frac{c}{f\sqrt{eff}} = \frac{3\times10^8}{3.6G\sqrt{2.19655}} = 0.0562$$

4.1.3.2 Calculations of input components length

a) Shunt Inductor (L_1)

$$L = \frac{\lambda_h \omega L_1}{2\pi Z_h} = \frac{(0.0566)(3.6G)(1.401n)}{130} = \overset{2.196mm}{\cancel{2.196mm}}\ \cancel{0.002196mm}$$

b) Series Capacitor (C_1)

$$L = \frac{\lambda_L \omega C_1 Z_L}{2\pi} = (0.0562)(3.6G)(0.926P)(25) = \overset{4.684mm}{\cancel{4.684mm}}\ \cancel{0.004684m}$$

4.1.3.3 Calculations of output components length

c) Shunt Capacitor (C_2)

Due to the impractical length resulting from the excessively large inductance value of C2 = 624.4 × 10^6 nH, L1 was substituted instead to achieve a feasible microstrip line length. This adjustment ensures the design remains physically realizable while maintaining the desired electrical performance.

$$L = \frac{\lambda_L \omega C_1 Z_L}{2\pi} = (0.0562)(3.6G)(0.926P)(25) = \overset{4.684\,mm}{\cancel{4.684\,mm}\ \cancel{0.004684\,mm}}$$

d) Series Inductor (L_2)

$$L = \frac{\lambda_h \omega L_2}{2\pi\ 2h} = \frac{(0.0566)(3.6G)(0.147n)}{130} = \overset{\cancel{0.2304\,mm}}{\cancel{0.2304\,mm}\ \cancel{0.0002304\,mm}}$$

4.1.4 Distributed Elements Before Optimization

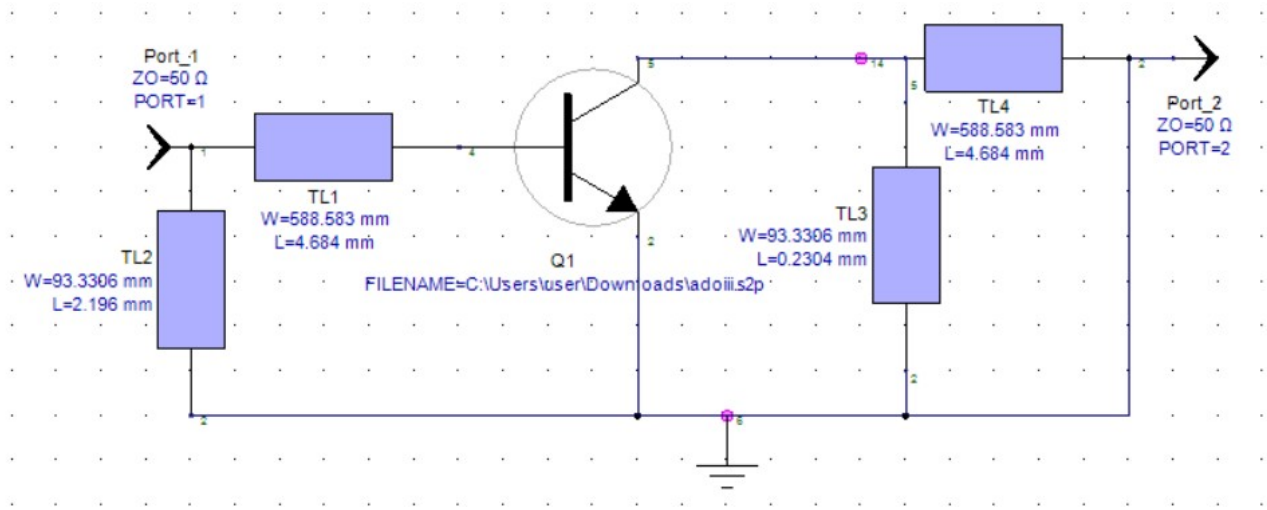


Figure 19: Distributed Elements Before Optimized

Figure 19 illustrates the distributed implementation of the LNA, where all lumped reactive components are replaced by equivalent microstrip transmission lines while preserving the original input and output matching topology. The distributed elements are implemented using microstrip lines labelled TL1 to TL4, corresponding to the capacitive and inductive elements of the lumped circuit.

At the input side, TL1 and TL2 form the distributed input matching network. TL1 represents the capacitive element and is implemented with a microstrip width of 588.583 mm and a physical length of 4.684 mm. TL2 corresponds to the inductive element, realized using a narrower microstrip width of 93.3306 mm and a length of 2.196 mm. These dimensions are selected to reproduce the desired capacitive and inductive behaviour at the target frequency of 3.6 GHz.

Similarly, the output matching network consists of TL3 and TL4. TL3 represents the inductive element and shares the same microstrip width of 93.3306 mm, with a physical length of 0.2304 mm. TL4 corresponds to the capacitive element and is implemented with a width of 588.583 mm and a length of 4.684 mm. Using identical widths for components of the same reactive type ensures consistent electromagnetic behaviour across the distributed matching network.

Overall, the distributed design accurately emulates the original lumped-element circuit while accounting for wave propagation effects at microwave frequencies. The calculated microstrip dimensions allow the distributed elements to exhibit equivalent reactance at 3.6 GHz, ensuring proper impedance matching, stable operation, and reliable gain and noise performance in the final LNA implementation.

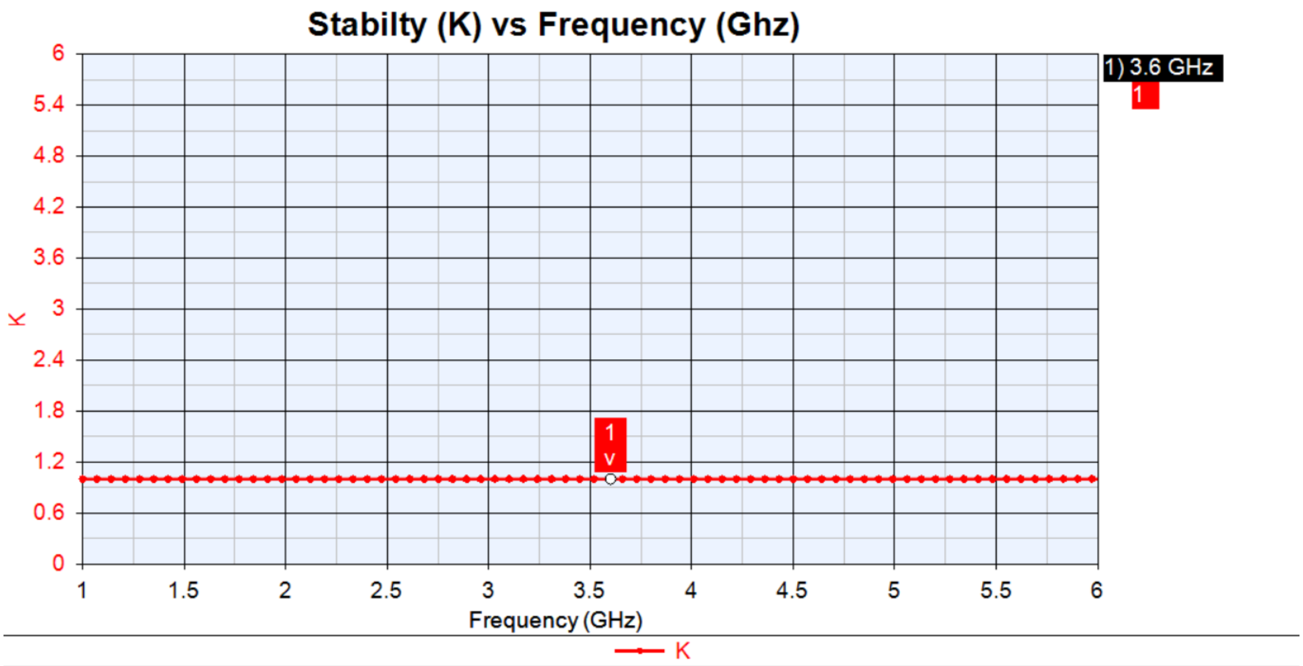


Figure 20: Distributed Stability Graph Before Optimized

Figure 20 illustrates the stability factor (K) of the distributed-element LNA before optimization. At the operating frequency of 3.6 GHz, the circuit exhibits a stability factor of approximately K = 1, indicating marginal stability. This condition suggests that the amplifier is neither clearly unconditionally stable nor unstable, and its behavior is highly sensitive to variations in source and load impedances, component tolerances, and layout parasitic. While the circuit can operate without oscillation under ideal conditions, the limited stability margin implies that further refinement is necessary to ensure robust performance across the intended frequency band. As a result, optimization is required to increase the stability factor beyond unity while preserving acceptable gain, impedance matching, and noise performance in the distributed design.

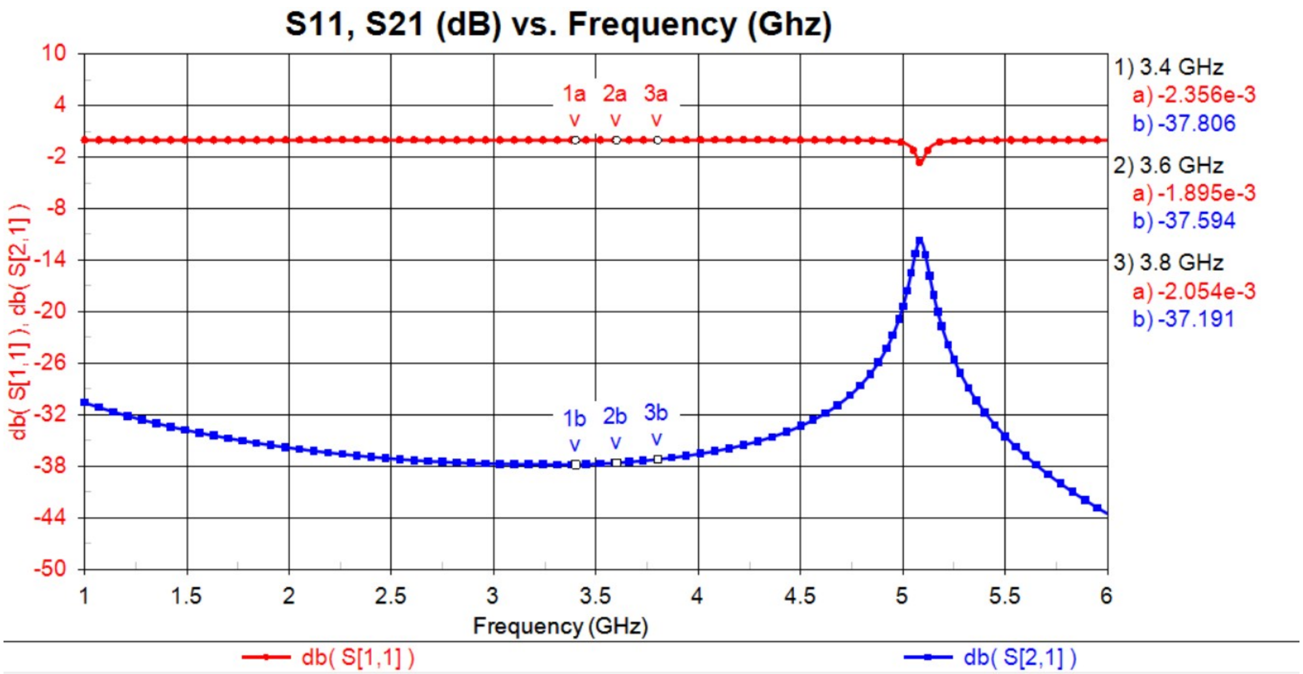


Figure 21: Distributed Gain Graph Before Optimized

Figure 21 presents the gain response of the distributed-element LNA before optimization. At the operating frequency of 3.6 GHz, the forward gain is approximately −37.594 dB, indicating that the circuit exhibits significant attenuation rather than amplification. This behaviour suggests that the distributed matching network is not yet providing the appropriate impedance transformation required for effective power transfer between the source, transistor, and load.

The low gain can be attributed to several factors, including inaccurate initial transmission-line dimensions, excessive insertion loss introduced by the distributed elements, and mismatch between the transistor's optimal source and load impedances. In addition, wave propagation effects such as phase cancellation and parasitic reactances become more pronounced in distributed circuits and can severely degrade gain if not properly accounted for. These results confirm that the distributed design, in its pre-optimization state, requires further tuning to align the matching networks with the intended operating frequency and to restore the expected amplification performance of the QPL9057 transistor.

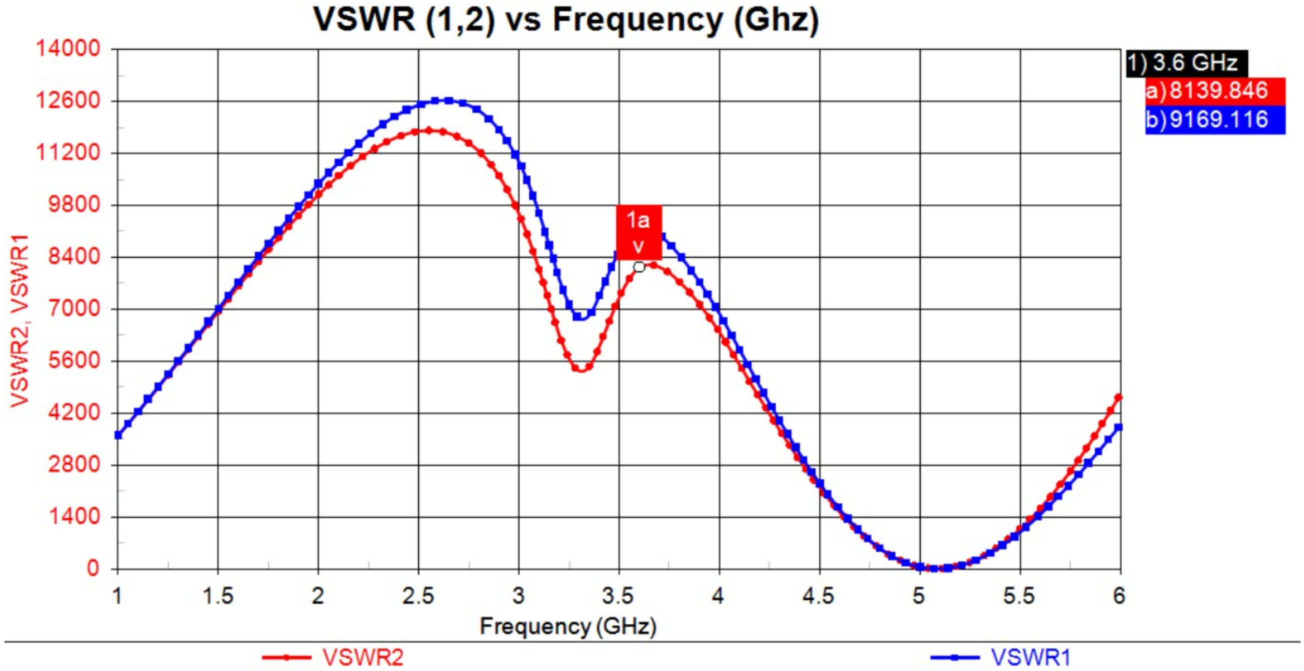


Figure 22: Distributed VSWR Before Optimized

Figure 22 illustrates the VSWR performance of the distributed-element LNA prior to optimization. At the operating frequency of 3.6 GHz, the input VSWR ($VSWR_1$) reaches a value of approximately 9169.116, while the output VSWR ($VSWR_2$) is around 8139.846. These extremely high VSWR values indicate severe impedance mismatch at both the input and output ports of the distributed circuit.

Such pronounced mismatching suggests that the initial transmission-line dimensions do not yet provide the required impedance transformation at the target frequency. As a result, a significant portion of the incident signal is reflected rather than transferred through the amplifier, leading to poor power delivery and degraded overall performance. Additionally, wave propagation effects, parasitic reactances, and frequency-sensitive behaviour inherent in distributed implementations can further intensify mismatch if not accurately modelled during the initial design stage.

The observed VSWR results confirm that the distributed circuit, prior to optimization, is far from an optimal matching condition. Therefore, further refinement of the transmission-line parameters is necessary to significantly reduce reflections, improve impedance matching, and enable proper amplification in subsequent optimization stages.

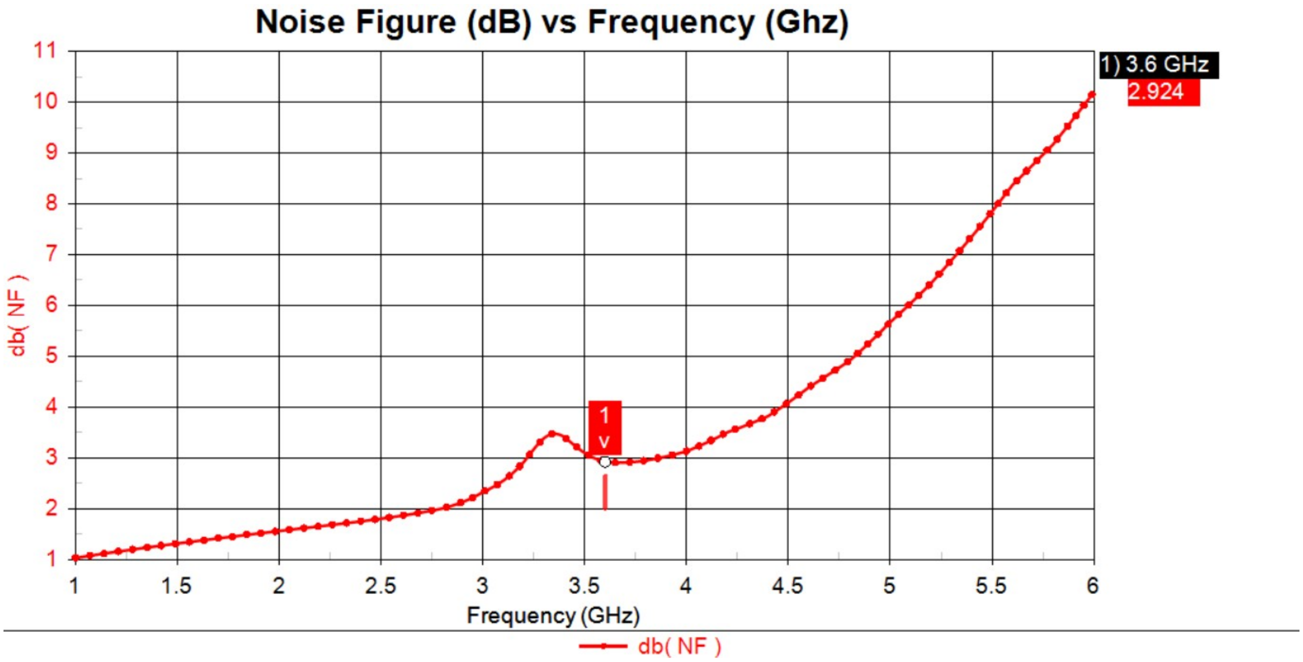


Figure 23: Distributed Noise Figure Graph Before Optimized

Figure 23 shows the noise figure performance of the distributed-element LNA prior to optimization. At the operating frequency of 3.6 GHz, the noise figure is approximately 2.924 dB. This value indicates that the distributed implementation introduces a moderate level of noise compared to the lumped-element design, which is expected in a pre-optimization stage where transmission-line dimensions and impedance matching have not yet been fully refined.

The increased noise figure can be attributed to severe impedance mismatches at the input, as reflected by the high VSWR values, which prevent the source from being presented with the optimal noise-matching condition. Additionally, losses introduced by the distributed transmission lines and substrate contribute to the overall noise performance. These results highlight the need for further optimization to improve input matching and reduce losses, thereby lowering the noise figure while maintaining stable operation at the target frequency.

4.1.5 Distributed Elements After Optimization

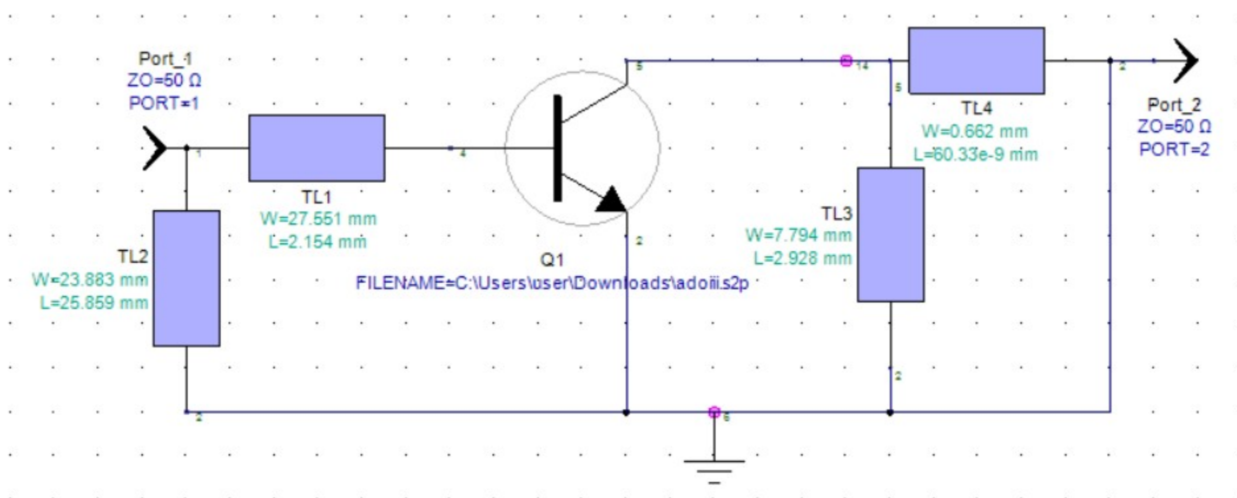


Figure 24: Distributed Elements After Optimized

Figure 24 shows the optimized distributed-element implementation of the LNA, where the initial transmission-line dimensions have been refined to improve impedance matching and overall circuit behavior at the target frequency of 3.6 GHz. The optimized design preserves the same circuit topology as the pre-optimization distributed model, while the physical dimensions of the microstrip lines have been adjusted to better emulate the required reactive elements.

At the input side, the matching network is formed by TL1 and TL2. TL1 is implemented with a microstrip width of 27.561 mm and a length of 2.154 mm, while TL2 is realized with a width of 23.883 mm and a length of 25.859 mm. These optimized dimensions allow the input impedance to be transformed closer to the desired source impedance of the transistor, improving power transfer from the 50-Ω input port.

On the output side, the matching network consists of TL3 and TL4. TL3 has a width of 7.794 mm and a length of 2.928 mm, whereas TL4 is implemented with a narrower width of 0.862 mm and a longer length of approximately 60.336 mm. This configuration enables more effective impedance transformation toward the 50-Ω load at the output port.

Overall, the optimized distributed circuit demonstrates a more realistic and physically consistent implementation of the matching networks, accounting for wave propagation effects at microwave frequencies. The refined microstrip dimensions provide improved impedance control and form the basis for enhanced stability, gain, VSWR, and noise performance in the subsequent simulation results.

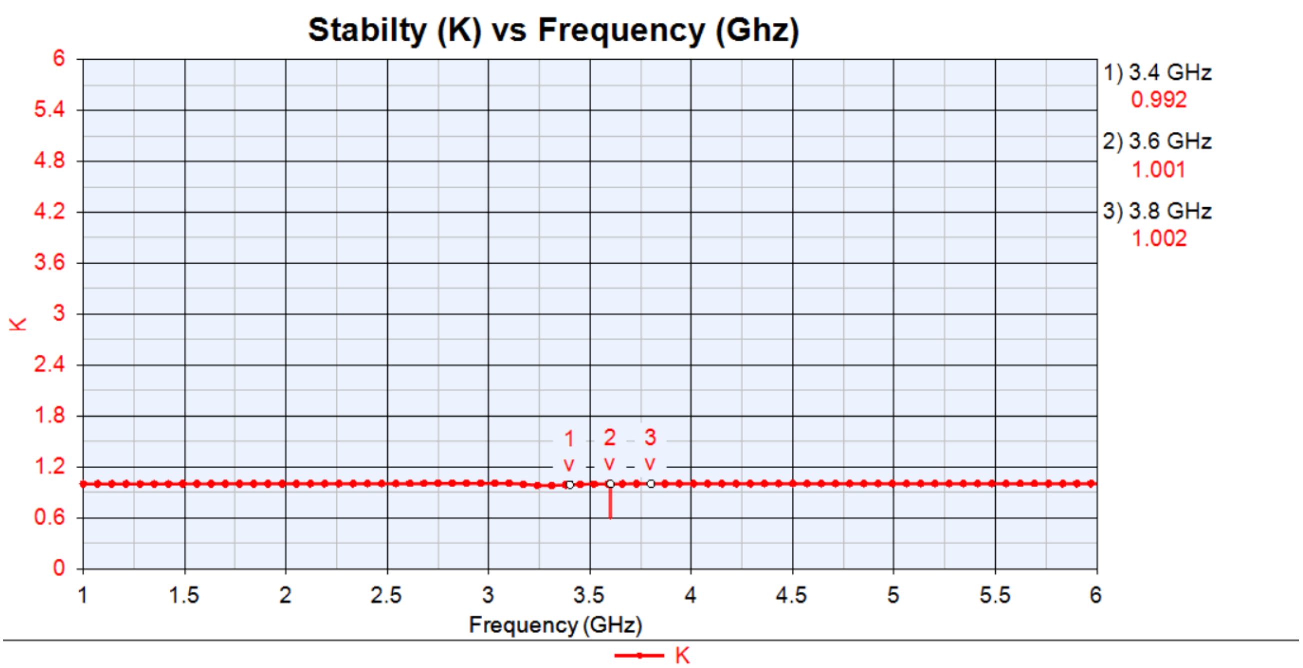


Figure 25: Distributed Stability Graph After Optimized

Figure 25 presents the stability factor (K) of the optimized distributed-element LNA. At the target operating frequency of 3.6 GHz, the circuit achieves a stability factor of approximately K = 1.001. This value indicates that the amplifier has transitioned from marginal stability to unconditional stability, as the stability factor slightly exceeds unity.

Although the stability margin is small, the result confirms that the optimized matching networks successfully improve the circuit's stability without introducing excessive damping. The proximity of K to 1 suggests that the design is efficiently balanced, maintaining stable operation while avoiding unnecessary degradation of gain or noise performance. However, due to the limited stability margin, the circuit remains sensitive to variations in component dimensions and fabrication tolerances, highlighting the importance of precise layout control in the distributed implementation.

Overall, the post-optimization stability result demonstrates that the applied transmission-line adjustments effectively enhance stability at 3.6 GHz, providing a suitable operating condition for the LNA within the intended 5G frequency band.

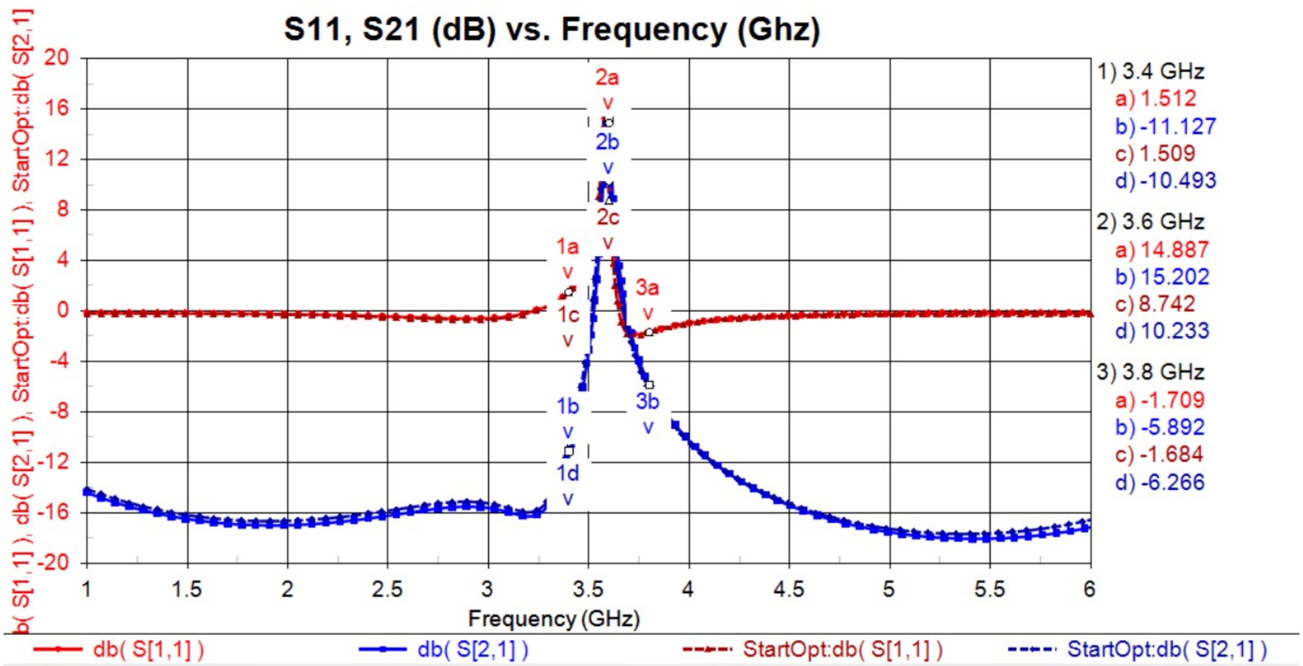


Figure 26: Distributed Gain Graph After Optimized

Figure X illustrates the forward gain performance of the optimized distributed-element LNA. At the target operating frequency of 3.6 GHz, the amplifier achieves a forward gain of approximately 15.202 dB. This result represents a substantial improvement compared to the pre-optimization distributed design, which exhibited significant attenuation instead of amplification.

The increase in gain confirms that the optimized transmission-line matching networks effectively transform the source and load impedances toward the optimal operating points of the QPL9057 transistor. By properly accounting for wave propagation effects and adjusting the physical dimensions of the microstrip lines, the distributed implementation enables efficient power transfer through the amplifier. The achieved gain level is appropriate for a low-noise amplifier operating in the sub-6 GHz 5G band and demonstrates the success of the optimization process in restoring and enhancing the amplification capability of the circuit.

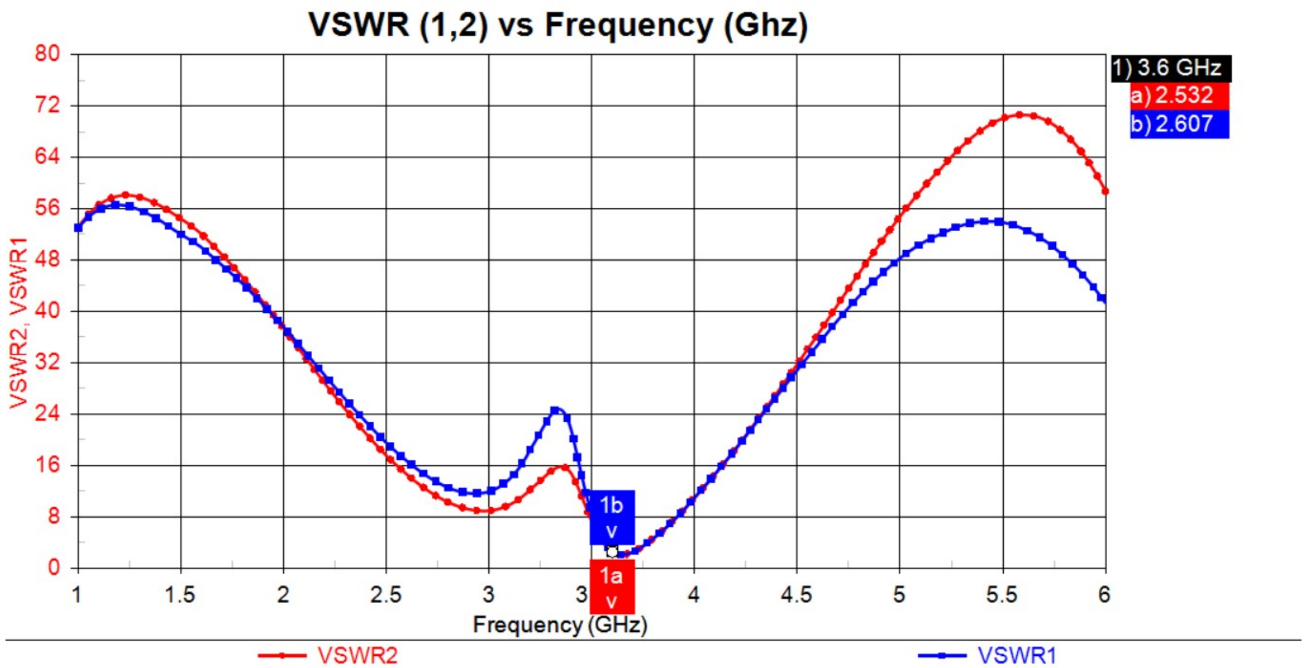


Figure 27: Distributed VSWR Graph After Optimized

Figure 27 illustrates the Voltage Standing Wave Ratio (VSWR) of the optimized distributed-element LNA at the operating frequency of 3.6 GHz. After optimization, the input VSWR ($VSWR_1$) is reduced to approximately 2.607, while the output VSWR ($VSWR_2$) is around 2.532. Compared to the extremely high VSWR values observed before optimization, these results indicate a substantial improvement in impedance matching at both ports.

Although the VSWR values are still higher than the ideal value of 1, they demonstrate that the optimized matching networks significantly reduce signal reflections and enhance power transfer between the source, the transistor, and the load. The remaining mismatch can be attributed to the limited degrees of freedom in the distributed matching network and the sensitivity of microstrip dimensions at microwave frequencies. Further fine-tuning or additional matching stages could potentially reduce the VSWR further; however, the achieved values are acceptable for a first optimized distributed implementation and confirm the effectiveness of the applied optimization process.

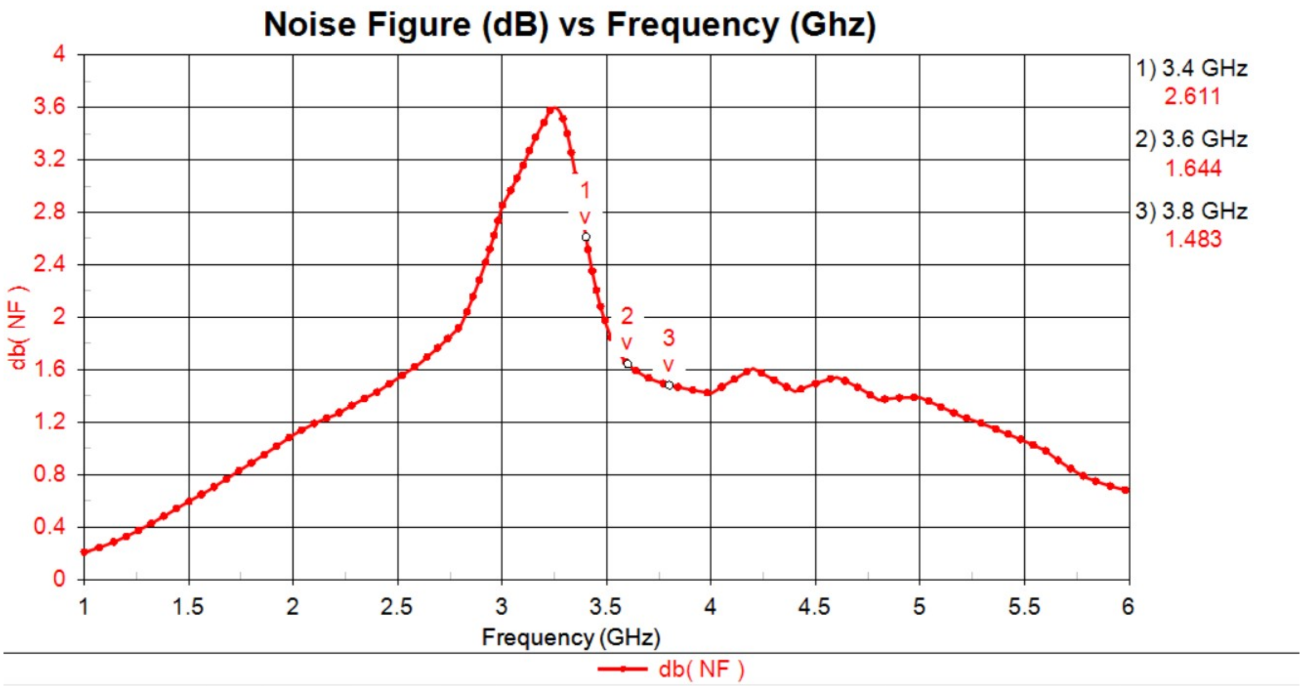


Figure 28: Distributed Noise Figure Graph After Optimized

Figure 28 presents the noise figure performance of the optimized distributed-element LNA at the operating frequency of 3.6 GHz. After optimization, the noise figure is reduced to approximately 1.644 dB, representing a significant improvement compared to the pre-optimization distributed design. This reduction indicates that the optimized matching networks are more closely aligned with the transistor's optimal noise-matching condition.

The improvement in noise figure can be attributed to better impedance matching at the input, which allows the source to be presented with an impedance closer to the optimum noise impedance of the transistor. In addition, the refinement of transmission-line dimensions helps minimize unnecessary losses within the distributed matching network. Overall, the achieved noise figure demonstrates that the optimization process effectively enhances noise performance while maintaining stable operation at the target frequency.

4.2 Circuit Layout

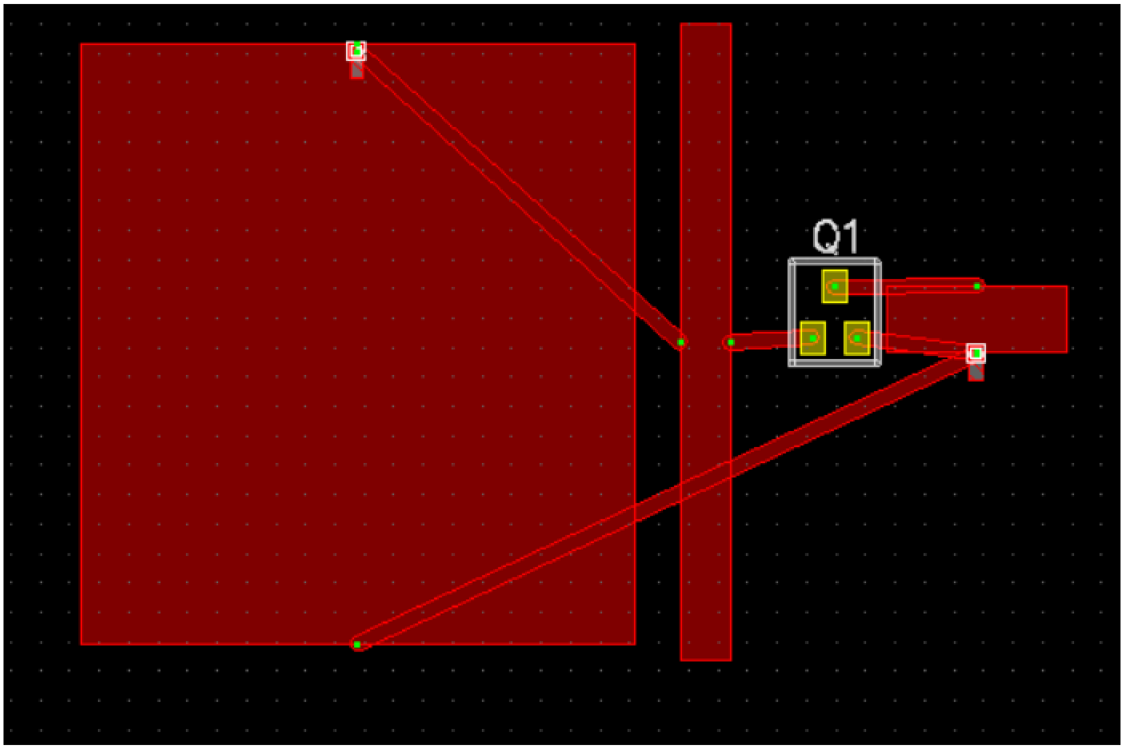


Figure 29: Layout

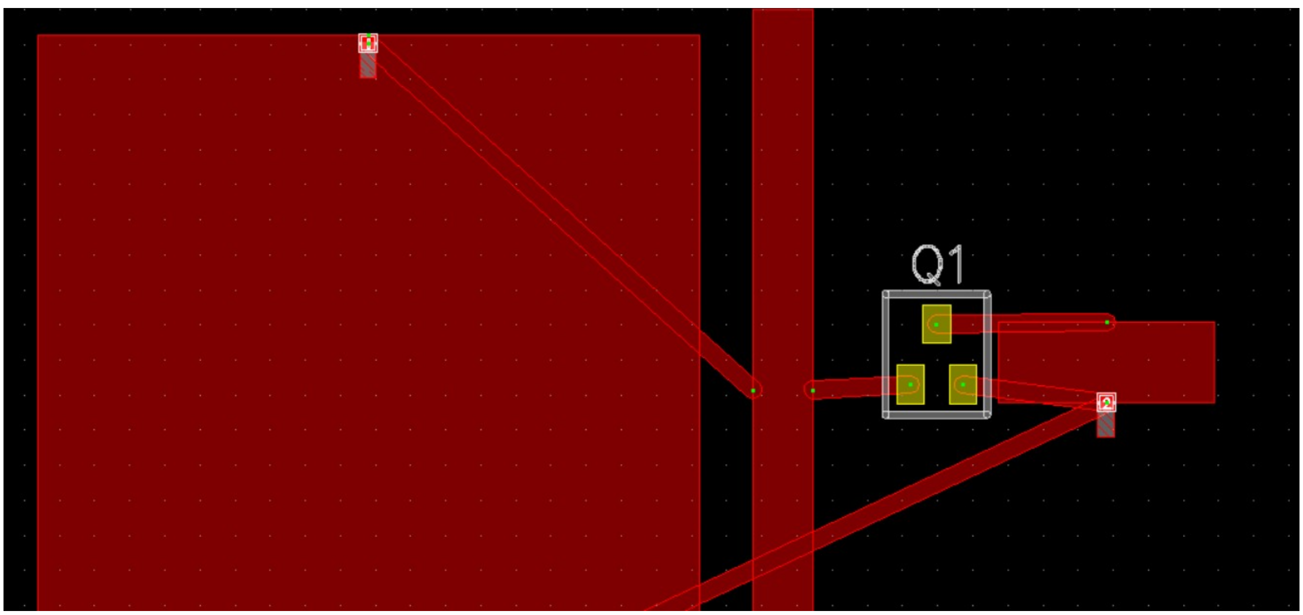


Figure 30: Close Up Layout

Figures 29 and 30 illustrate the final physical layout of the optimized distributed LNA designed to operate at 3.6 GHz. The layout translates the optimized schematic into a microstrip implementation while preserving the intended electrical behavior. The active device, Q1, is positioned near the center of the layout to minimize interconnection lengths and to maintain a clear separation between the input and output matching networks.

A notable feature of the layout is the large ground plane region, which provides a stable RF reference and helps suppress unwanted radiation and surface current propagation. This is particularly important for low-noise amplifier operation, as effective grounding directly influences stability and noise performance. The vertical ground strip located close to the transistor further enhances grounding effectiveness and contributes to isolation between the input and output sections.

The layout is implemented on Rogers RT/duroid 5880, which is well suited for high-frequency applications. Its low dielectric constant ($\varepsilon r = 2.2$) allows wider microstrip lines, reducing conductor loss and improving fabrication tolerance, while the very low loss tangent ($\tan \delta = 0.0009$) minimizes dielectric losses at 3.6 GHz. These properties help maintain accurate impedance control and preserve signal integrity, even with relatively long transmission-line sections.

The input and output microstrip lines are routed diagonally and linearly toward their respective ports to achieve the required electrical lengths for the distributed matching networks. Although this routing increases the overall footprint, it is necessary to realize the desired inductive and capacitive behavior at the target frequency. The high-frequency performance of the Rogers RT/duroid 5880 substrate helps compensate for these spatial requirements by limiting parasitic losses and unwanted coupling effects.

While the layout is electrically functional and meets the optimized performance targets, it is not fully optimized from a layout efficiency and electromagnetic robustness perspective. The long diagonal traces can introduce additional parasitic inductance and increase sensitivity to electromagnetic coupling. Furthermore, the limited use of dense via stitching along the ground edges and near the transistor source may reduce the effectiveness of the RF ground return path. In a practical fabricated design, these factors could slightly affect stability margins and noise performance.

To further improve the layout, additional ground vias should be placed near the transistor source and along the edges of critical microstrip lines to reduce ground inductance. Where possible, the long diagonal routing can be shortened or re-oriented to minimize parasitic effects. Implementing via fences or grounded guard structures between the input and output sections would further enhance isolation, stability, and overall layout robustness. With these refinements, the layout would be better suited for reliable and repeatable performance in sub-6 GHz 5G LNA applications.

5.0 CONCLUSION

This project successfully demonstrates the design and optimization of a 3.6 GHz low-noise amplifier (LNA) for sub-6 GHz 5G applications using the QPL9057 transistor. The main goal of the work was to achieve a practical balance between important RF performance parameters, including gain, noise figure, stability, and impedance matching. Through systematic analysis and iterative optimization, the final distributed design achieved a forward gain of 15.202 dB, a low noise figure of 1.644 dB, acceptable VSWR at both input and output ports, and stable operation with a stability factor slightly above unity at the target frequency.

Several challenges were encountered during the design process, particularly when converting the lumped-element matching networks into a distributed microstrip implementation. At microwave frequencies, small changes in transmission-line dimensions and layout can significantly affect circuit behavior. In addition, trade-offs between stability and noise performance had to be carefully managed to avoid degrading overall amplifier performance. These challenges highlighted the importance of practical optimization and layout-aware design rather than relying solely on theoretical calculations.

The choice of Rogers RT/duroid 5880 as the substrate material contributed significantly to the success of the design. Its low dielectric constant ($\varepsilon r = 2.2$) and very low loss tangent ($\tan \delta = 0.0009$) helped reduce signal loss, maintain consistent impedance, and improve the reliability of the distributed matching networks at 3.6 GHz. This made the substrate well suited for the intended 5G application.

Overall, the results show that a well-optimized distributed LNA can meet the key performance requirements for sub-6 GHz 5G receiver front ends while accounting for real-world implementation constraints. For future work, further improvements could be achieved by performing full electromagnetic simulations of the layout, refining grounding and isolation techniques, and integrating the LNA with other RF front-end components to improve compactness and manufacturability.

Project/Assignment Evaluation Form (50%)

CO1: Demonstrate the scattering parameters of a microwave circuit. CO2: Convert lumped circuit elements into microstrip form.
CO3: Design microwave amplifier and microwave filter or oscillator, and verify one of the designs through fabrication.

Project/ Assignment:	**Students' Names:**
[redacted] Design	1. [redacted]
	2. AZWAN
	3.
	4.

PART A

PO2
Identify, formulate, research literature and analyze complex electronics engineering problems reaching substantiated conclusions.

Indicator (Domain)	**Complex Engineering Problem (WP/EA)**	**Marks Allocated**	**Marks (*M*)**					**Weighted Marks Obtained**
			1	**2**	**3**	**4**	**5**	
Identify and analyze detailed procedures for the calculations.	NA	5	Show minimal ability to identify and analyze detailed procedures for the calculations.	Show some ability to identify and analyze detailed procedures for the calculations.	Show adequate ability to identify and analyze detailed procedures for the calculations.	Show good ability to identify and analyze detailed procedures for the calculations.	Show excellent ability to identify and analyze detailed procedures for the calculations.	**(=M X 1)**
Formulate the parameters involved using appropriate calculations and according to the required design objectives.	NA	5	Show minimal ability to formulate the parameters involved using appropriate calculation and not according to design objectives	Show some ability to formulate the parameters involved using appropriate calculation and not according to design objectives	Show adequate ability to formulate the parameters involved using appropriate calculation and according to design objectives	Show good ability to formulate the parameters involved using appropriate calculation and according to design objectives	Show excellent ability to formulate the parameters involved using appropriate calculation and according to design objectives	**(=M X 1)**

Total (PART A)	**/10**

PART B

PO3 Ability to design solutions for complex electronics engineering problems with appropriate consideration for public health and safety, culture, society, and environment.								
Indicator (Domain)	**Complex Engineering Problem (WP/EA)**	**Marks Allocated**	**Marks (*M*)**					**Weighted Marks Obtained**
			1	**2**	**3**	**4**	**5**	
Analyze the lumped-to-distributed element transformation scheme relevant to the design.	NA	10	Show minimal ability to analyze the lumped-to-distributed element transformation scheme.	Show some ability to analyze the lumped-to-distributed element transformation scheme.	Show adequate ability to analyze the lumped-to-distributed element transformation scheme.	Show good ability to analyze the lumped-to-distributed element transformation scheme.	Show excellent ability to analyze the lumped-to-distributed element transformation scheme.	**(=Mx2)**
Analyze the transmission line theory and/or formulae involved relevant to the design.	NA	7.5	Show minimal ability to analyze the knowledge in transmission line theory and/or formulae involved.	Show some ability to analyze the knowledge and need to include transmission line theory and/or formulae involved.	Show adequate ability to analyze the knowledge and the need to include transmission line theory and/or formulas involved with minimal analysis.	Show good ability to analyze the knowledge and need to include transmission line theory and/or formulae involved with adequate analysis.	Show excellent ability to analyze the knowledge and need to include transmission line theory and/or formulae involved with comprehensive analysis.	**(=Mx1.5)**
Evaluate the design with reliable explanation that suits technological application	NA	2.5	Failed to evaluate the design with no explanation that suits the technological application.	Evaluate the design with an unreliable explanation that suits the technological application.	Evaluate the design with a reliable explanation that suits the technological application.	Evaluate the design with a good and reliable explanation that suits the technological application.	Evaluate the design with a comprehensive and reliable explanation that suits the technological application.	**(=Mx0.5)**

Total (PART B)	**/20**

PART C

PO5 Ability to create, select, and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modeling, involving complex electronics engineering problems.								
Indicator (Domain)	**Complex Engineering Problem (WP/EA)**	**Marks Allocated**	**Marks (*M*)**					**Weighted Marks Obtained**
			1	**2**	**3**	**4**	**5**	
Explain the simulated design based on the calculations.	WP1, WP2, WP3	2.5	Show minimal ability to explain the simulated design based on the calculations.	Show some ability to explain the simulated design based on the calculations.	Show adequate ability to explain the simulated design based on the calculations.	Show a good ability to explain the simulated design based on the calculations.	Show an excellent ability to explain the simulated design based on the calculations.	**(=Mx0.5)**
Design the circuit layout according to the design objectives using modern engineering and IT tools with/without guidance .	WP1, WP2, WP3	7.5	Show minimal ability to design the circuit layout according to the design objectives using modern engineering and IT tools with full guidance.	Show some ability to design the circuit layout according to the design objectives using modern engineering and IT tools with full guidance.	Show adequate ability to design the circuit layout according to the design objectives using modern engineering and IT tools with minimal guidance.	Show a good ability to design the circuit layout according to the design objectives using modern engineering and IT tools without guidance.	Show an excellent ability to design the circuit layout according to the design objectives using modern engineering and IT tools without guidance.	**(=Mx1.5)**
Specify and demonstrate the design with parametric evaluations.	WP1, WP2, WP3	7.5	Show minimal ability to specify and demonstrate the design with parametric evaluations.	Show some ability to specify and demonstrate the design with parametric evaluations.	Show adequate ability to specify and demonstrate the design with parametric evaluations.	Show a good ability to specify and demonstrate the design with parametric evaluations.	Show an excellent ability to specify and demonstrate the design with comprehensive parametric evaluations.	**(=Mx1.5)**

Adapt any deviation of the actual result from the target specification and include suggestions on how to mitigate it.	WP1, WP2, WP3	2.5	Show minimal adaptation towards the deviation in the results with required target specification and minimal suggestion to mitigate the design problem.	Show some adaptation towards the deviation in the results with required target specification and some suggestion to mitigate the design problem.	Show adequate adaptation towards the deviation in the results with required target specification and adequate suggestion to mitigate the design problem.	Show good adaptation towards the deviation in the results with required target specification and good suggestion to mitigate the design problem.	Show excellent adaptation towards the deviation in the results with required target specification and excellent suggestion to mitigate the design problem.	**(=Mx0.5)**
							Total (PART C)	**/20**

TOTAL MARKS (PART A+PART B+PART C)	**/50**